

Statics
 (At rest)

Dynamics

Kinematics
 (Independent of
 cause of motion)

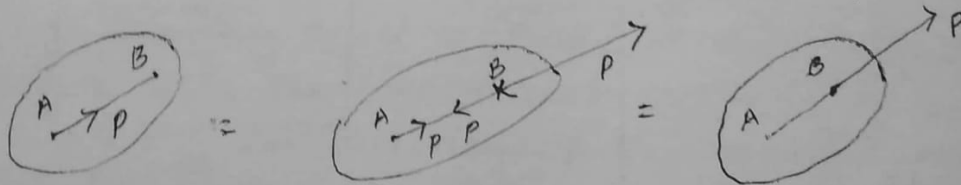
Kinetics
 (Dependent on
 cause of
 motion)

Basic Laws of Motions:

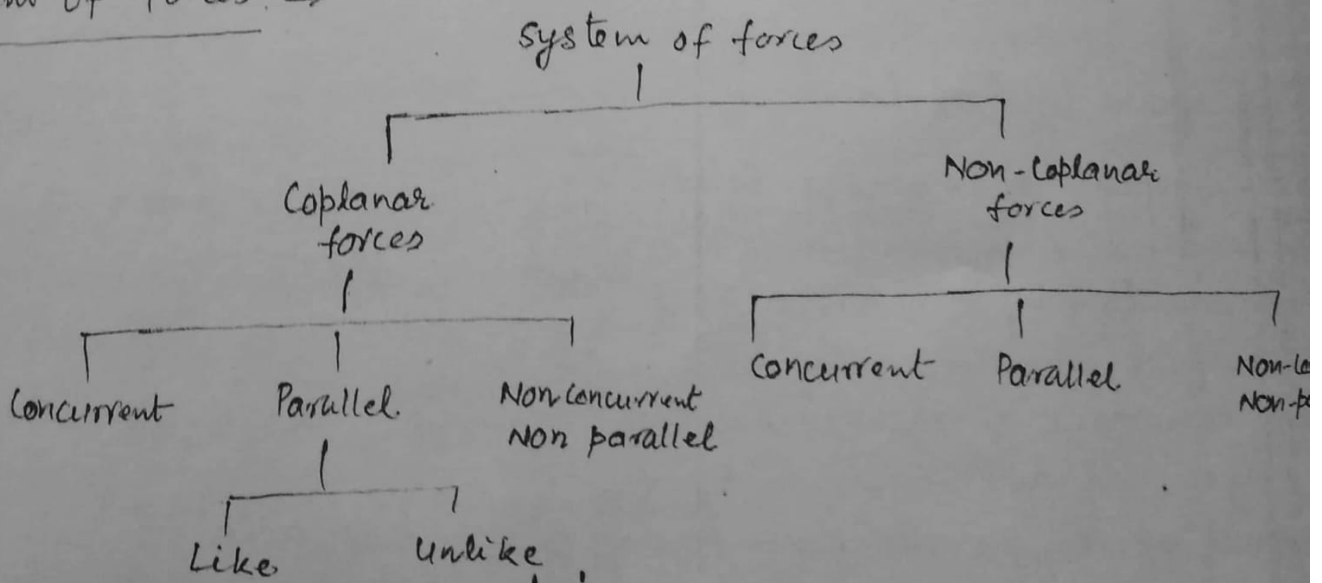
- 1) Newton's First Law
- 2) Newton's Second Law
- 3) Newton's Third Law
- 4) Law of Gravitation

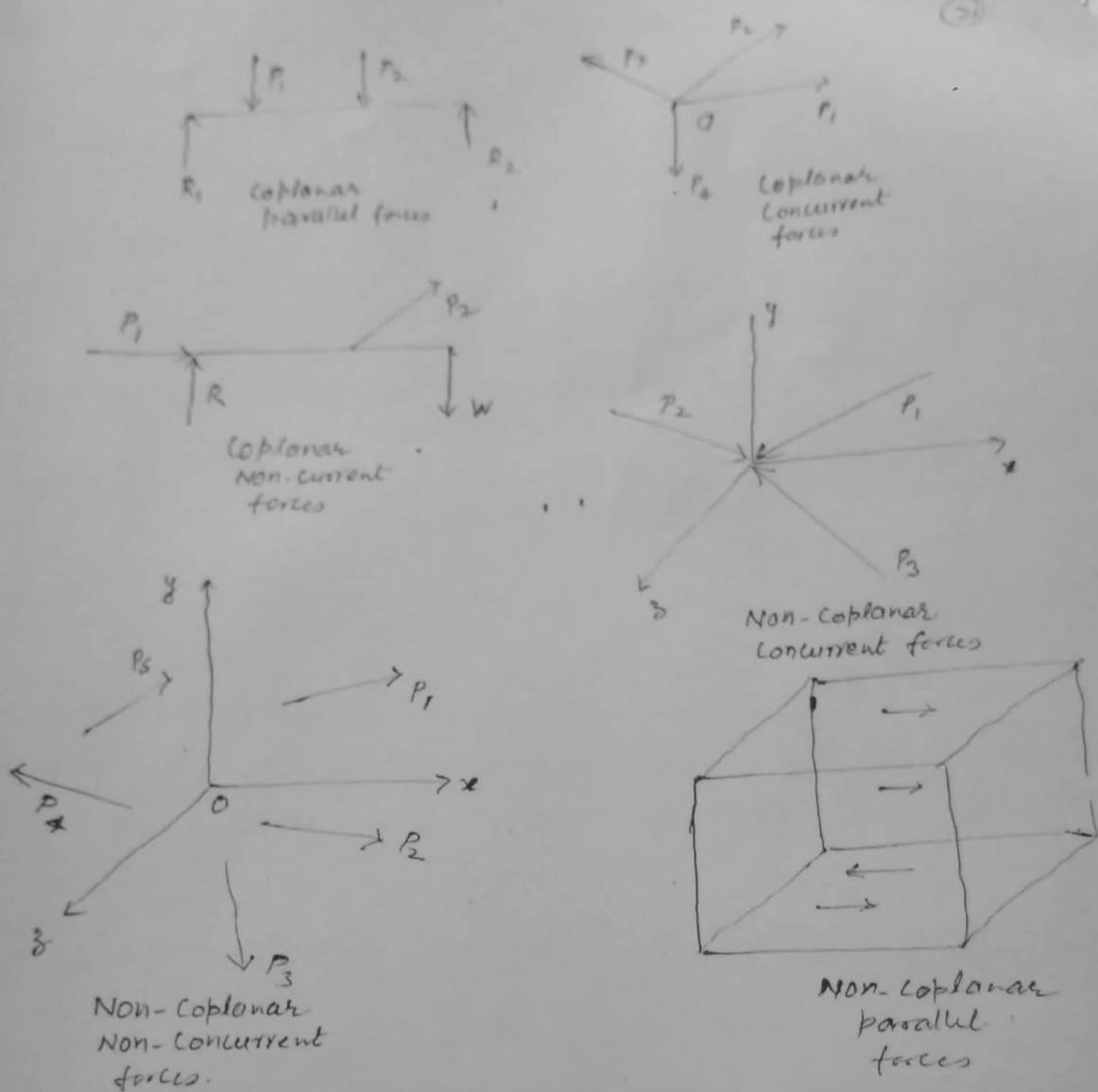
$$F = \frac{G m_1 m_2}{r^2} \quad \text{Where } G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Principle of transmissibility: → When the point of application of a force acting on a body is shifted to any point on the line of action of the force without changing its direction, there occurs no change in the equilibrium state of the body.



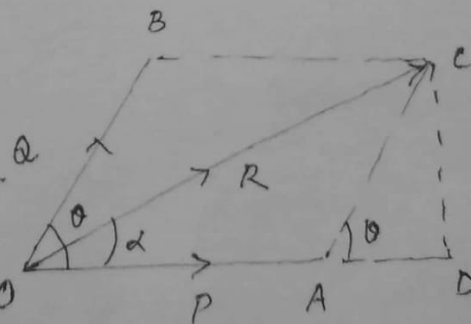
System of forces: →





Triangle Rule of forces:

If two forces, acting at a point are represented in magnitude and direction by the two adjacent sides of a parallelogram, then their resultant is represented in magnitude and direction by the diagonal of the parallelogram passing through that point.



$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

Q. No. 1: The resultant of two forces P and Q acting at a point R. The resultant R gets doubled when Q is either doubled or its direction is reversed. Show that P, Q and R conform to the ratio $P:Q:R = \sqrt{2}:\sqrt{3}:\sqrt{2}$ (2)

Solⁿ:

$$R^2 = P^2 + Q^2 + 2PQ \cos \theta \quad \text{--- (1)}$$

When Q is doubled,

$$(2R)^2 = P^2 + (2Q)^2 + 2P(2Q) \cos \theta$$

$$4R^2 = P^2 + 4Q^2 + 4PQ \cos \theta \quad \text{--- (2)}$$

When Q is reversed in direction

$$(2R)^2 = P^2 + (-Q)^2 + 2P(-Q) \cos \theta$$

$$4R^2 = P^2 + Q^2 - 2PQ \cos \theta \quad \text{--- (3)}$$

Solving (1) and (3)

$$5R^2 = 2P^2 + 2Q^2 \quad \text{--- (4)}$$

Solving (2) and (3)

$$4R^2 = P^2 + 2Q^2 \quad \text{--- (5)}$$

$$P:Q:R = \sqrt{2}:\sqrt{3}:\sqrt{2}$$

Q. No. 2: → Two forces equal to P and 2P respectively act on a particle. When the first force is increased by 120 N and second force is doubled, the direction of the resultant remains the same. Determine the value of P.

Solⁿ: $\tan \alpha = \frac{2P \sin \theta}{P + 2P \cos \theta}$ --- (1), θ is the angle betⁿ two given forces

When forces get changed

$$\tan \alpha = \frac{4P \sin \theta}{(P + 120) + 4P \cos \theta} \quad \text{--- (2)}$$

From (1) and (2)

$$\frac{2P \sin \theta}{P + 2P \cos \theta} = \frac{4P \sin \theta}{(P + 120) + 4P \cos \theta} \Rightarrow P = 120 \text{ N.}$$

Q-No. 3: → The resultant of two forces P and Q is right angle to P . Show that angle betⁿ the forces is $\cos^{-1}(-P/Q)$. (4)

Solⁿ:

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

Since $\alpha = 90^\circ$

$$\tan 90^\circ = \frac{Q \sin \theta}{P + Q \cos \theta}$$

$$\alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

$$P + Q \cos \theta = 0$$

$$\theta = \cos^{-1}(-P/Q)$$

NO. 4: → Two forces one of which is double the other has resultant of 260 N. If the direction of the larger force is reversed and the other remains unaltered, the resultant reduces to 180 N. Determine the magnitude of the force and angle betⁿ the forces

solⁿ: →

$$R^2 = P^2 + Q^2 + 2PQ \cos \theta$$

$$(260)^2 = P^2 + (2P)^2 + 2P(2P) \cos \theta$$

$$= P^2 + 4P^2 + 4P^2 \cos \theta$$

$$= 5P^2 + 4P^2 \cos \theta \quad \text{--- (1)}$$

and

$$(180)^2 = P^2 + (-2P)^2 + 2P(-2P) \cos \theta$$

$$(180)^2 = 5P^2 - 4P^2 \cos \theta \quad \text{--- (2)}$$

Solving (1) and (2)

$$P = 100 \text{ N}$$

$$Q = 2P = 200 \text{ N}$$

From (1)

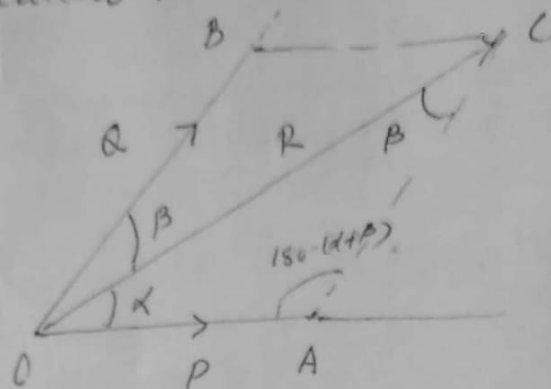
$$260^2 = 5(100)^2 + 4(100)^2 \cos \theta$$

$$\cos \theta = 0.44$$

$$\theta = 63.89^\circ$$

Resolution of forces:

Finding the components of a given force in two given direction is called resolution.



Applying Sine rule,

$$\frac{OA}{\sin \beta} = \frac{AC}{\sin \alpha} = \frac{OC}{\sin \{180 - (\alpha + \beta)\}}$$

$$\frac{P}{\sin \beta} = \frac{Q}{\sin \alpha} = \frac{R}{\sin (\alpha + \beta)}$$

$$P = R \frac{\sin \beta}{\sin (\alpha + \beta)}$$

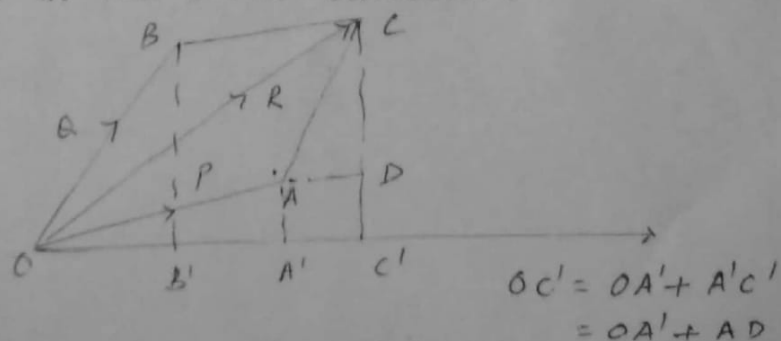
$$Q = R \frac{\sin \alpha}{\sin (\alpha + \beta)}$$

When the force R is to be resolved along perpendicular directions,
 $\alpha + \beta = 90^\circ$

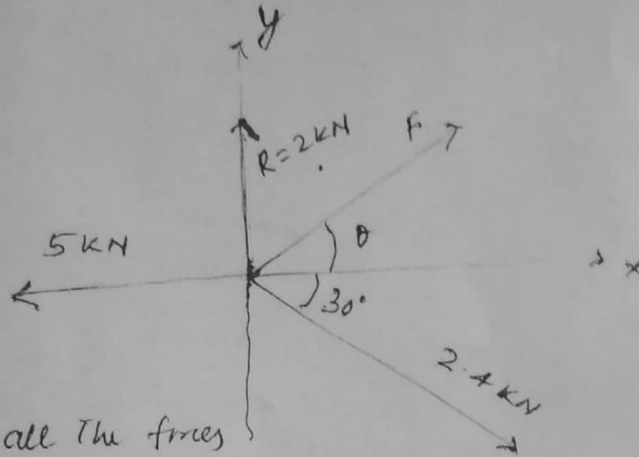
$$P = R \frac{\sin (90 - \alpha)}{\sin 90} \Rightarrow P = R \cos \alpha$$

$$Q = R \frac{\sin \alpha}{\sin 90} = R \sin \alpha$$

Theorem of resolved parts : $\rightarrow$ The algebraic sum of the resolved parts of two forces in a given direction is equal to the resolved part of their resultant in the same direction.



Q. No. 1 :→ The force system shown in the figure has a resultant of 2 kN acting up along the y-axis. Make calculations for the force F both in Magnitude & Direction.



Solⁿ:
Resolving all the forces in vertical directions

$$F \sin \theta - 2.4 \sin 30^\circ = 2$$

$$F \sin \theta = 3.2 \text{ kN} \quad \text{--- (1)}$$

Resolving all forces in horizontal direction

$$F \cos \theta + 2.4 \cos 30^\circ - 5 = 0$$

$$F \cos \theta = 2.922 \text{ kN} \quad \text{--- (2)}$$

Squaring (1) and (2) then adding

$$F = \sqrt{(3.2)^2 + (2.922)^2} = 4.33 \text{ kN}$$

$$\textcircled{1} \div \textcircled{2}$$

$$\tan \theta = \frac{3.2}{2.922} = 1.095$$

$$\boxed{\theta = 47.6^\circ}$$

(Coplanar forces)

Resultant of Coplanar Concurrent forces →

Resultant $R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$

and its inclination θ to x-axis is given by

$$\tan \theta = \frac{\sum F_y}{\sum F_x}$$

The algebraic sum of the resolved parts of two forces in a given direction is equal to the resolved part of their resultant in the same direction.

Equilibrium of a particle →

A particle will be in equilibrium when resultant of all the forces acting on it is zero.

$$R = 0 \Rightarrow \sum F_x = 0$$

$$\sum F_y = 0$$

Equilibrium of two force-body:



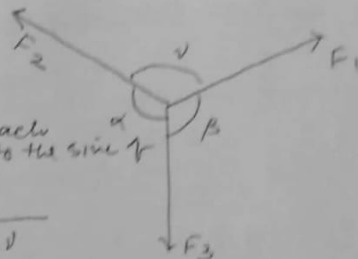
Two forces must be collinear and must have same Magnitude but opposite direction.

Equilibrium of a three-force body

Lami's theorem:

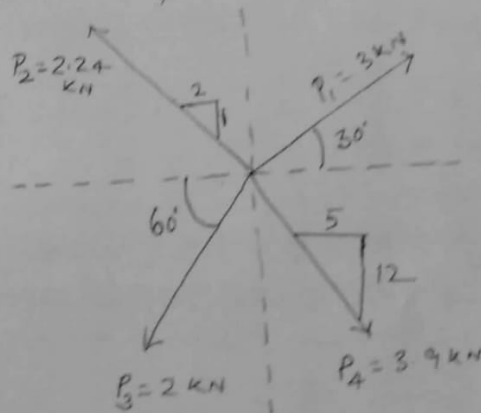
If three forces acting on a particle are in eqn then each force is directly proportional to the sine of angle betⁿ other two forces.

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$



For the equilibrium of three force body, acting forces must be concurrent, and must form a closed triangle.

Q.No. 1 → Determine the resultant, both in Magnitude and direction, of the four forces acting on a body as shown in the figure.



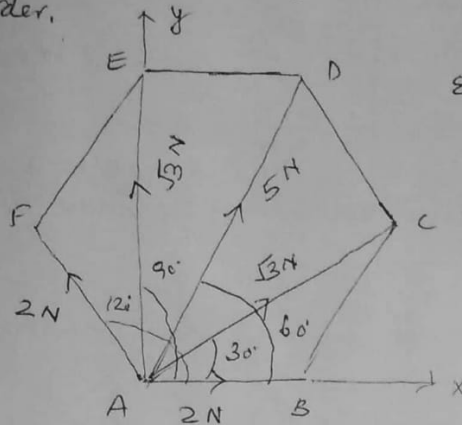
$$\sum F_x = 1.094 \text{ kN}$$

$$\sum F_y = -2.83 \text{ kN}$$

$$R = 3.034 \text{ kN}$$

$$\alpha = \tan^{-1} \left(\frac{-2.83}{1.094} \right) = 68.86^\circ$$

Q. NO. 2: Find the resultant of forces $2, \sqrt{3}, 5, \sqrt{3}$ and $2N$ that act at an angular point of a regular hexagon towards the other angular points taken in order.



$$\Sigma F_x = 2 + \sqrt{3}\cos 30^\circ + 5\cos 60^\circ + \sqrt{3}\cos 90^\circ + 2\cos 120^\circ$$

$$= 5N$$

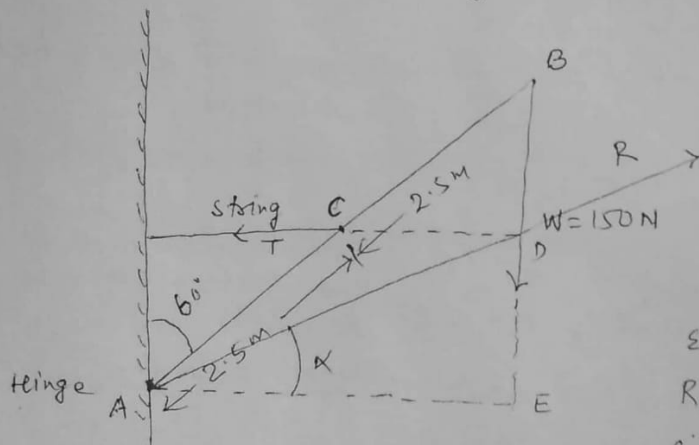
$$\Sigma F_y = \sqrt{3}\sin 30^\circ + 5\sin 60^\circ + \sqrt{3}\sin 90^\circ + 2\sin 120^\circ$$

$$= 5\sqrt{3}N$$

$$R = \sqrt{5^2 + (5\sqrt{3})^2} = 10N$$

$$\tan \alpha = \frac{5\sqrt{3}}{5} \Rightarrow \alpha = 60^\circ$$

NO. 3: A rigid prismatic bar 5m long is supported in a vertical plane by a hinge at one end and by a horizontal string attached to it at its midpoint. A 150N weight hangs vertically from the free end of the bar. Neglecting weight of the bar, determine tension in the string and direction of the reaction at the hinge.



$$BD = BC \cos 60^\circ = 2.5 \times \frac{1}{2} = 1.25m = DE$$

$$AE = AB \sin 60^\circ = 4.33N$$

$$\tan \alpha = \frac{1.25}{4.33} \Rightarrow \alpha = 16.1^\circ$$

$$\Sigma F_x = 0$$

$$R \cos \alpha - T = 0 \Rightarrow R \cos \alpha = T \quad \text{--- (1)}$$

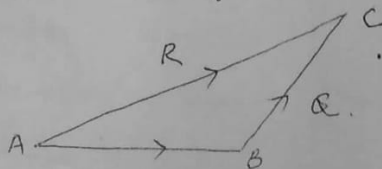
$$\Sigma F_y = 0$$

$$R \sin \alpha - 150 = 0 \Rightarrow R \sin \alpha = 150 \quad \text{--- (2)}$$

Solving (1) and (2)

$$T = 519.57N$$

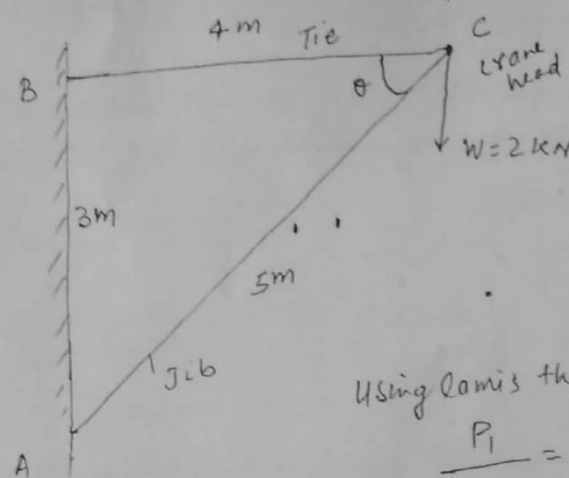
Triangle law of forces: If the two forces acting simultaneously on a body are represented by sides of a triangle taken in order, their resultant is represented by the closing side of the triangle taken in opposite order.



is also often stated as follows.

If a system of three forces acting upon a body can be represented in magnitude and direction by the sides of a triangle, taken in order, then system will be in equilibrium.

Q. NO. 1 In a jib crane, the jib and tie rod are 5 m and 4 m long respectively. The height of the crane post is 3 m, and the tie rod remains horizontal. Determine the forces produced in the jib and the tie rod when a load of 2 kN is suspended at the crane head.



$$\sin \theta = \frac{3}{5} \Rightarrow \theta = 36.87^\circ$$

Using Lami's theorem

$$\frac{P_1}{\sin 270^\circ} = \frac{P_2}{\sin 53.13^\circ} = \frac{2}{\sin 36.87^\circ}$$

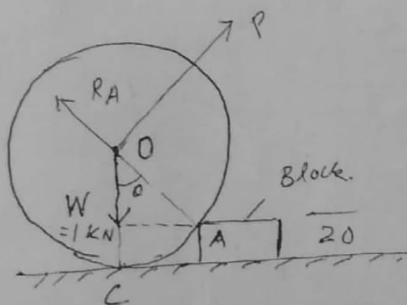
$$P_1 = -3.33 \text{ kN (Comp.)}$$

$$P_2 = 2.667 \text{ kN (Tension)}$$

Q. NO. 2 $\rightarrow$ A uniform wheel of 50 cm diameter and 1 kN weight rests against a rigid rectangular block of thickness 20 cm. Considering all the surfaces smooth, determine

(a) Least pull to be applied through the centre of wheel to just turn it over the corner of the block.

(b) Reaction of the block.



$$\cos \theta = \frac{OB}{OA} = \frac{50-20}{50} \Rightarrow \cos \theta = 0.6 \Rightarrow \theta = 53.13^\circ$$

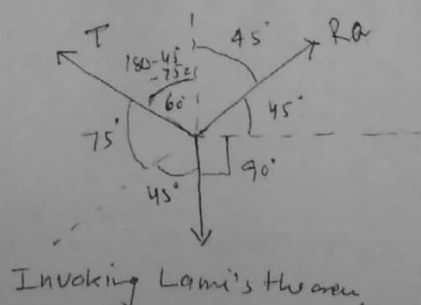
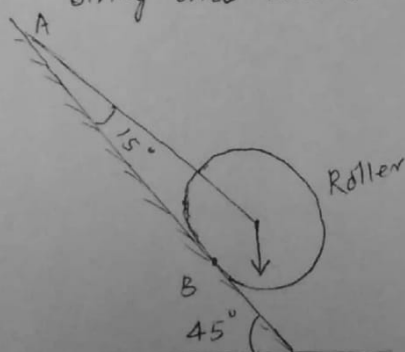
$$\frac{W}{\sin 90^\circ} = \frac{P}{\sin 126.87^\circ} = \frac{R_A}{\sin 143.13^\circ}$$

Pull P must be applied normal to OA if it is to be minimum.

$$P = 0.8 \text{ kN}$$

$$R_A = 0.6 \text{ kN}$$

Q. NO. 3 $\rightarrow$ A roller of weight 500 N rests on a smooth inclined plane and is kept free from rolling down by a string as shown in fig. Work out tension in the string and reaction at point of contact B.



Invoking Lami's theorem,

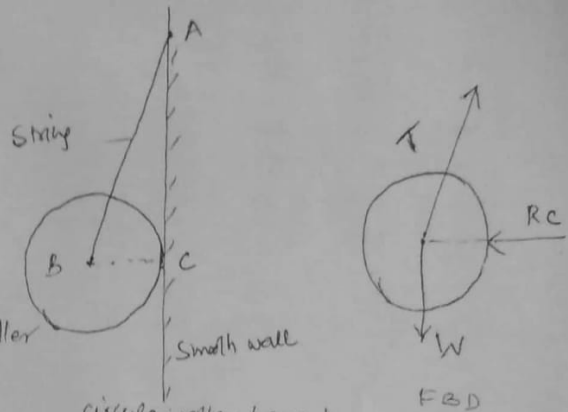
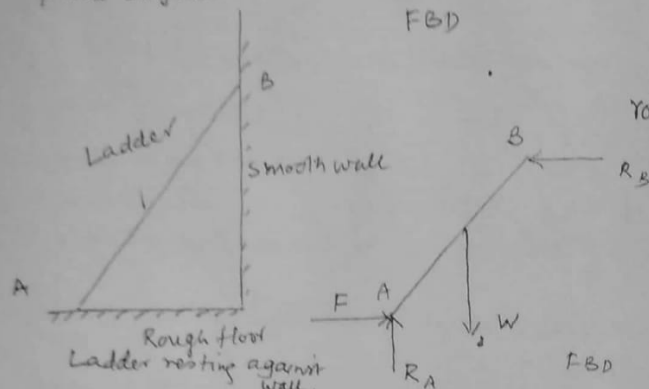
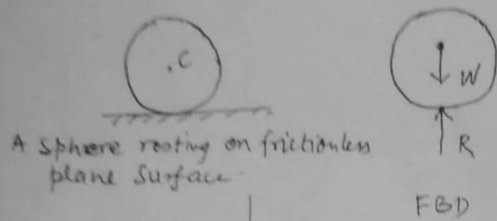
$$\frac{W}{\sin(60+45)} = \frac{R_B}{\sin(75+45)} = \frac{T}{\sin 90}$$

$$R_B = 448.24 \text{ N}$$

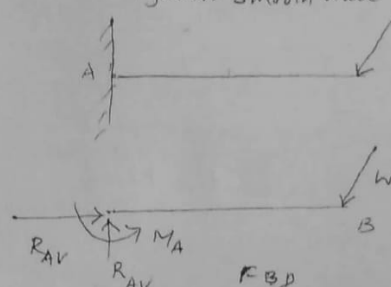
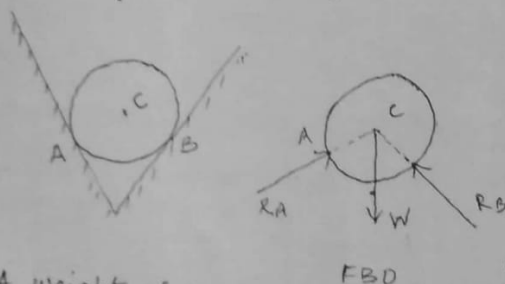
$$T = 365.94 \text{ N}$$

Free-body diagram: Force analysis of a structure is made in a simplified way by considering the equilibrium of a portion of the structure. For that, the portion is drawn separately showing applied forces, self weight and reactions at the point of contacts with other bodies. The resulting diagram is known as free body diagram (FBD).

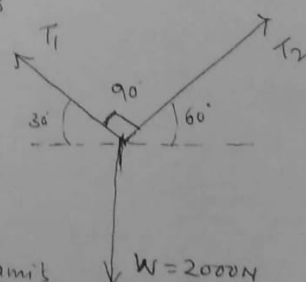
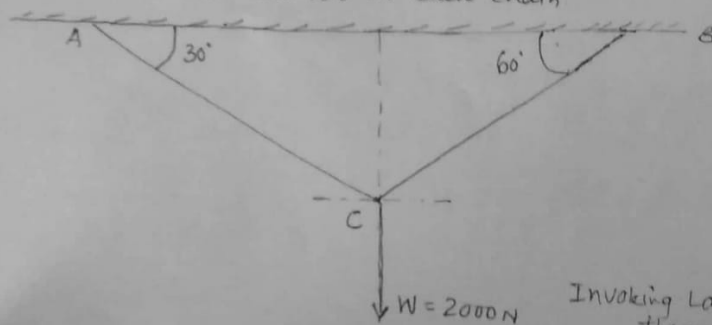
In free body diagram, all supports, like walls, floors, hinges etc. are removed and replaced by the reactions which these supports exert on the body.



A Sphere resting in a V-Shaped groove



NO. A weight of 2000 N is supported by two chains AC and BC as shown in fig. Determine the tension in each chain.



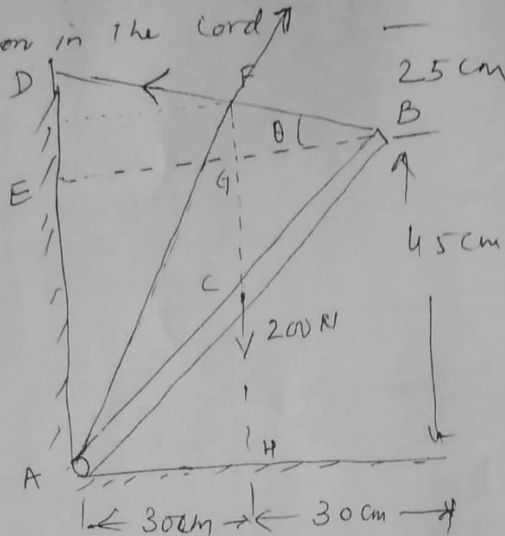
Invoking Lami's theorem

$$\frac{T_1}{\sin 150} = \frac{T_2}{\sin 120} = \frac{2000}{\sin 90}$$

$$T_1 = 1000\text{ N}$$

$$T_2 = 1732\text{ N}$$

Q NO1: A rod AB of weight 200N is supported by a cable BD and the corner of wall and floor surface as shown in fig. Find the reaction at A and tension in the cord.



$$\tan \theta = \frac{BE}{ED} = \frac{60}{25}$$

$$\frac{F_G}{30} = \tan \theta \Rightarrow F_G = 30 \times \frac{60}{25}$$

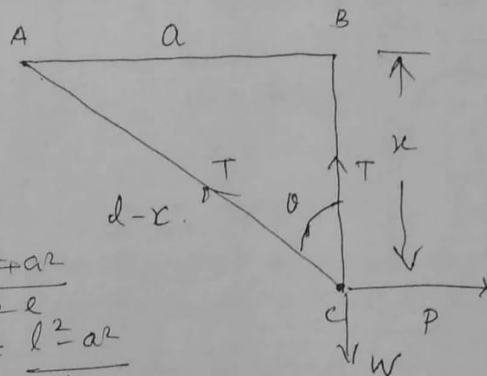
$$F_H = F_G + F_G = 45 + F_G$$

$$\frac{T}{\sin 152.5^\circ} = \frac{R}{\sin 112.6^\circ} = \frac{200}{\sin \{360 - \{152.5^\circ + 94.9^\circ\}\}}$$

$$T = 93.23 \text{ N}$$

$$R = 186.51 \text{ N}$$

Q NO2: A string of length l is fastened to two points A and B at the same level at a distance a apart. A ring of weight w can slide on the string and a horizontal force p is applied to it such that it is in equilibrium vertically below B. Show that $p = \frac{wa}{l}$ and that the tension in the string is $\frac{w(l^2 + a^2)}{2l^2}$.



$$(l-x)^2 = x^2 + a^2$$

$$x = \frac{l^2 - a^2}{2l}$$

$$l-x = l - \frac{l^2 - a^2}{2l} = \frac{l^2 + a^2}{2l}$$

$$\cos \theta = \frac{BC}{AC} = \frac{x}{l-x} = \frac{l^2 - a^2}{l^2 + a^2}$$

$$\sin \theta = \frac{AB}{l-x} = \frac{a \times 2l}{l^2 + a^2} = \frac{2al}{l^2 + a^2}$$

$$\sum F_x = 0$$

$$p = T \sin \theta$$

$$\sum F_y = 0$$

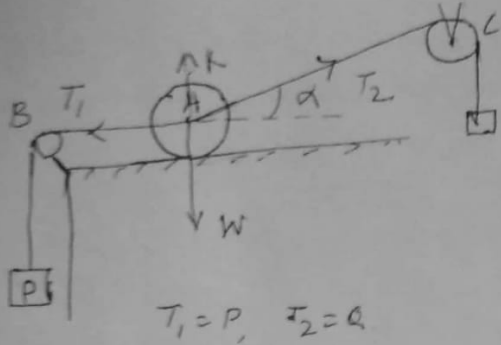
$$T + T \cos \theta = w$$

$$T \left(1 + \frac{l^2 - a^2}{l^2 + a^2} \right) = w$$

$$T = \frac{w(l^2 + a^2)}{2l^2}$$

$$p = \frac{w(l^2 + a^2)}{2l^2} \times \frac{2al}{l^2 + a^2} = \frac{wa}{l}$$

Not



A ball of weight W rest upon a smooth horizontal surface and has attached to its centre two strings AB and AC which pass over frictionless pulley B and C and carry load P and Q as shown in fig. If the string AB is horizontal, Find the angle α which the string AC makes with the horizontal when the ball is in a position of equilibrium. Also determine the pressure betⁿ ball and the plane

Resolving forces horizontally

$$T_1 = T_2 \cos \alpha$$

$$P = Q \cos \alpha$$

Resolving forces vertically (R is reaction force)

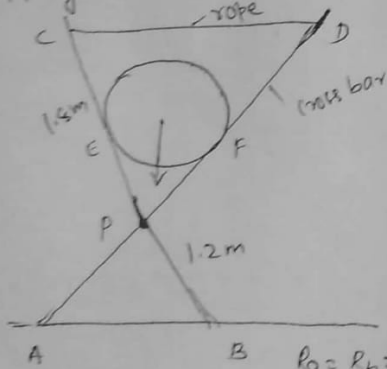
$$R = W - T_2 \sin \alpha$$

$$= W - Q \sqrt{1 - \cos^2 \alpha}$$

$$= W - Q \sqrt{1 - \frac{P^2}{Q^2}}$$

$$R = W - \sqrt{Q^2 - P^2}$$

Example: A steel cylinder 1m in diameter and weighing 500N is placed between two cross arms which makes an angle of 60° with each other and are pinned at P as shown in fig. A rope attached to the arms near their upper ends prevents them from slipping. Assume the surface of the floor to be smooth, determine the tension in the rope.



$$R_a = R_b = \frac{500}{2} = 250 \text{ N}$$

The cylinder is in eq^m.

$$\sum F_x = 0, \quad R_e \cos 30^\circ - R_f \cos 30^\circ, \quad R_e = R_f$$

$$\sum F_y = 0, \quad R_e \sin 30^\circ + R_f \sin 30^\circ - 500 = 0$$

$$0.5 R_e + R_f (0.5) = 500 \quad \left. \begin{array}{l} R_e = R_f \\ 0.5 R_e + R_f (0.5) = 500 \end{array} \right\} R_e = R_f = 500 \text{ N}$$

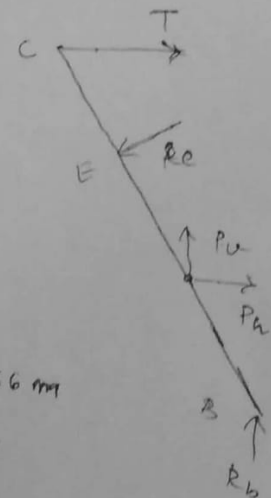
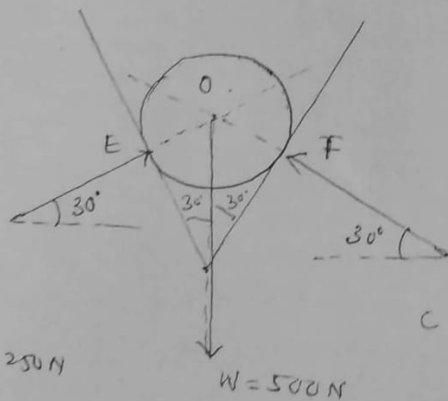
Considering equilibrium of arm BC, and taking Moment about point P

$$\sum M_P = 0, \quad T \times 1.8 \cos 30^\circ - R_e \times PE - R_b \times 1.2 \sin 30^\circ$$

$$\text{In } \triangle OEP, \quad \frac{OE}{PE} = \tan 30^\circ \Rightarrow PE = \frac{OE}{\tan 30^\circ} = \frac{0.5}{0.577} = 0.866 \text{ m}$$

$$T \times 1.8 \cos 30^\circ = R_e \times 0.866 + R_b \times 1.2 \sin 30^\circ$$

$$T = \frac{563}{1.557} = 373.95 \text{ N}$$



Q. No. A cylinder of weight W and radius r is in contact with a vertical wall at A and supported at B by a horizontal rope BC. The bar is hinged to the wall at A and supported at B by a horizontal rope BC. Find the value of the angle θ that the bar should make with the wall so that the tension in the rope is minimum. Assume frictionless conditions.



FBD of cylinder



$$\sum F_x = 0$$

$$R_f \sin \theta - W = 0$$

$$R_f = W / \sin \theta$$

$$\sum M_A = 0$$

$$T \times AC - R_f \times AF = 0$$

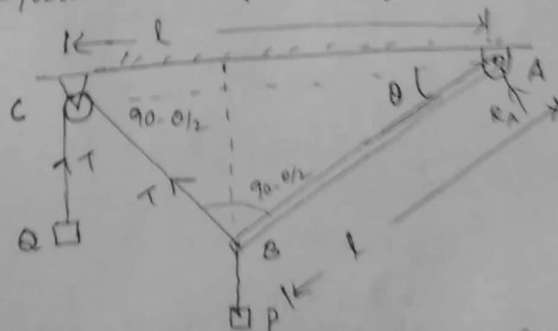
$$T = \frac{W \sin \theta \cos \theta / 2}{\sin \theta \cdot l \cos \theta}$$

$$\frac{dT}{d\theta} = 0 \Rightarrow \theta = 60^\circ$$

$$T_{\min} = \frac{4WR}{3}$$



Q. No. A uniform rod AB of negligible weight l is hinged at the end A and supported at end B by a string as shown. Find the value of angle θ corresponding to the position of equilibrium of bar if $Q = P/2$.



$$\sum M_A = 0$$

$$P(l \cos \theta - \frac{l}{2} \sin(90 - \theta)) = 0$$

$$P \cos \theta - \frac{l}{2} \cos \theta = 0$$

$$P(2 \cos^2 \theta - 1) - \frac{l}{2} \cos \theta = 0$$

$$2P \cos^2 \theta - \frac{l}{2} \cos \theta - P = 0$$

$$\cos \theta = \frac{Q \pm \sqrt{Q^2 + 4P^2}}{4P}$$

$$\text{Substituting } \frac{Q}{P} = \frac{1}{2}$$

$$\cos \theta = \frac{1}{4} \left(\frac{1}{2} \pm \sqrt{\frac{1}{4} + 4} \right)$$

$$= \frac{1}{8} (1 \pm 5.744)$$

$$\cos \theta = 0.843$$

$$\theta = 65^\circ$$

1. No. Two prismatic bars AB and CD are welded together in form of a rigid and suspended in a vertical plane as shown in figure. Determine the angle θ that the bar AB will make with vertical when a load of 100 N is applied at the end D. Two bars are identical and each weighs 50 N.

$$\sum M_A = 0$$

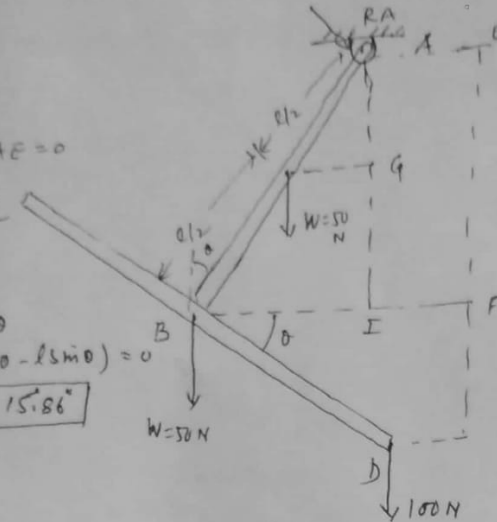
$$50 \times 4 + 50 \times 6 - 100 \times AE = 0$$

$$AE = IF$$

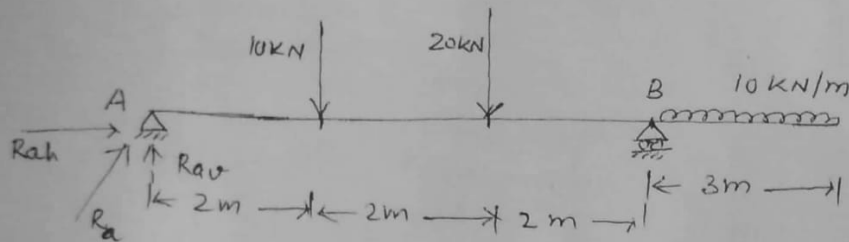
$$IF = 8F - 8I = 4/2 \cos \theta - 1 \sin \theta$$

$$50(4/2 \sin \theta) + 50 \times 1 \sin \theta - 100(4/2 \cos \theta - 1 \sin \theta) = 0$$

$$\tan \theta = \frac{5}{17.5} \Rightarrow \theta = 15.86^\circ$$



2. No. Determine the reactions at A and B



$$R_{ah} = 0$$

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M = 0$$

$$\sum M_A = 0$$

$$10 \times 2 + 20 \times 4 - R_b \times 6 + (10 \times 3) \times 7.5 = 0$$

$$R_b = 54.17 \text{ kN}$$

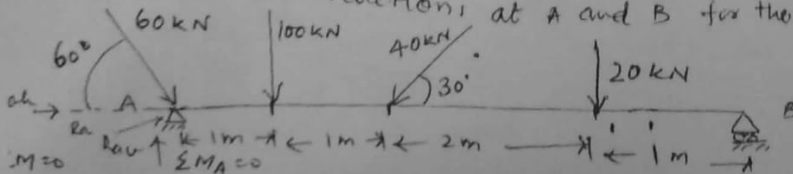
$$\sum F_y = 0$$

$$R_a - 10 - 20 + R_b - 10 \times 3 = 0$$

$$R_{av} = 5.83$$

$$R_a = \sqrt{R_{ah}^2 + R_{av}^2} = 0 + (5.83)^2 = 5.83 \text{ kN}$$

3. No. Determine the reaction at A and B for the beam loaded as shown in fig.



$$\sum M = 0$$

$$\sum F_x = 0$$

$$\sum F_y = 0$$

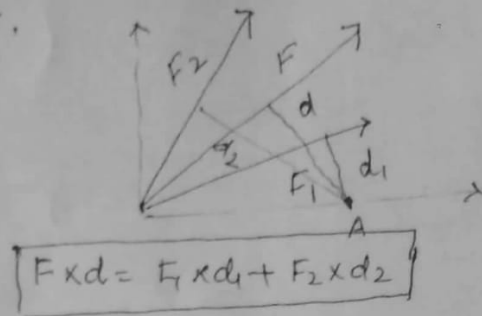
$$100 \times 1 + 40 \sin 30 \times 2 + 20 \times 4 - R_b \times 5 = 0 \Rightarrow R_b = 44 \text{ kN}$$

$$\sum F_x = 0 \quad R_{ah} + 60 \cos 60 - 40 \cos 30 = 0 \Rightarrow R_{ah} = 4.64 \text{ kN}$$

$$\sum F_y = 0 \quad R_{av} - 60 \sin 60 - 100 - 40 \sin 30 - 20 + R_b = 0 \Rightarrow R_{av} = 147.96 \text{ kN}$$

$$R_a = \sqrt{R_{ah}^2 + R_{av}^2} = 148 \text{ kN} \quad \theta = \tan^{-1} \left(\frac{R_{av}}{R_{ah}} \right) = \tan^{-1} \left(\frac{147.96}{4.64} \right) \Rightarrow \theta = 88.20^\circ$$

Variignon's theorem: (Law of Moments) It states that the moment of resultant of all the forces in a plane about any point is equal to the algebraic sum of moments of all the forces about the same point.



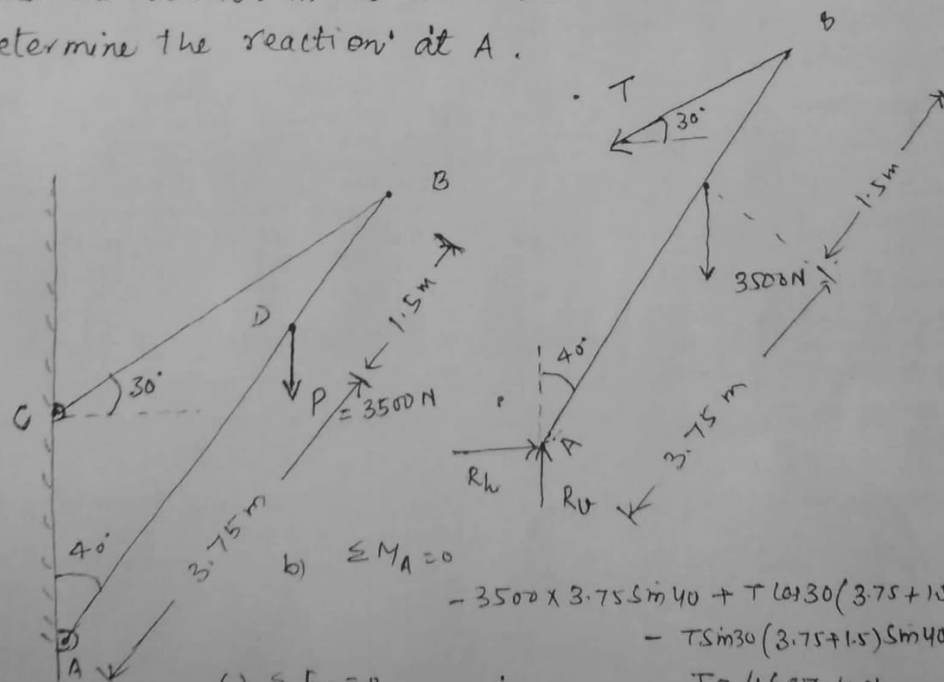
$$F \times d = F_1 \times d_1 + F_2 \times d_2$$

Principle of Moments: A body is acted upon by a number of forces (coplanar), will be in equilibrium, if the algebraic sum of moments of all the forces about a point lying in the same plane is zero.

Mathematically

$$\sum M = 0$$

- Q. No. A load P of 3500 N is acting on the boom which is held by a cable BC as shown in fig. Neglecting the weight of boom,
- Draw the free body diagram of the boom.
 - Find the tension in the cable BC .
 - Determine the reaction at A .



$$b) \sum M_A = 0$$

$$-3500 \times 3.75 \sin 40 + T \cos 30 (3.75 + 1.5) \cos 40$$

$$- T \sin 30 (3.75 + 1.5) \sin 40 = 0$$

$$T = 4697.4 \text{ N}$$

$$c) \sum F_x = 0$$

$$R_h - 4697.4 \cos 30 = 0$$

$$R_h = 4068.1 \text{ N}$$

$$\sum F_y = 0$$

$$R_v - 3500 - 4697.4 \sin 30 = 0$$

$$R = \sqrt{R_h^2 + R_v^2} = \sqrt{(4068.1)^2 + (8448.7)^2}$$

$$\theta = \tan^{-1} \left(\frac{R_v}{R_h} \right) = 7124.5^\circ$$

$$\theta = \tan^{-1} \left(\frac{R_v}{R_h} \right) = 55.2^\circ$$

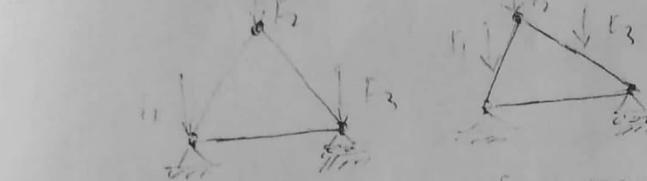
Analysis of Plane Trusses and Frames

Structure 4 System of connected members built to support or transfer loads acting on them and safely withstand these forces. These are broadly classed as 1) Trusses 2) Frames 3) Machines

(roofs of sheds at railway platform workshop, Industrial buildings, bridge trusses)

Truss → It is a system of uniform bars or members (of circular section, angle section or section etc.) joining together at their ends by rivelling or welding and constituted to support loads. Members of truss are straight members and loads are applied at joints. Every member of a truss is a two force member only, since they are either in tension or in compression.

These are used in covering long spans and rail road bridges.

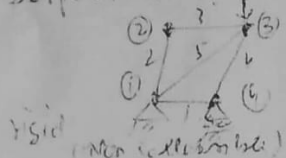


Perfect frame: A frame which is made of members just sufficient to keep it in equilibrium, when loaded, without change in its shape. Substructure is triangle.
 $m = 2j - 3$

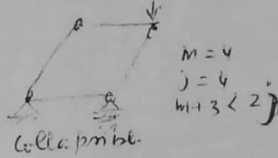
Imperfect frame: A frame which does not satisfy the equation $m = 2j - 3$

If $m > 2j - 3$ (Redundant frame)
 If $m < 2j - 3$ deficient frame (Collapsible)

Rigid or perfect truss: (Non collapsible) when external forces are removed



$m = 5$
 $j = 4$
 $m + 3 = 2j$



$m = 4$
 $j = 4$
 $m + 3 < 2j$

If $m + 3 > 2j$ it means that truss contains more members than reqd to be just rigid and is over rigid and statically indeterminate.
 If $m + 3 < 2j$ it means that contains less members than required to be just rigid and is collapsible or under rigid.

Statically determinate: If the eqⁿ of static eq^m alone are sufficient to determine axial forces in the members without considering their deformation.

Basic Assumption for perfect truss: Joints of truss are pin connections and frictionless.

Assumption for forces in members of a perfect frame

1. All members are pin jointed
2. The frame is loaded at the joints
3. The frame is a perfect one
4. Weights of members, unless stated otherwise, is regarded as negligible.

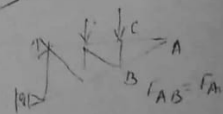
(Normally two of supports (one hinged other roller))

Forces in the members may be found out either by analytical method or graphic method.

Analytical Method for the forces (Method of truss analysis)

1.) Method of joints

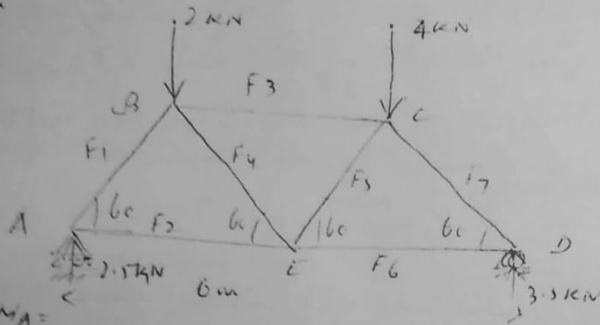
(Before member) 2.) Method of sections (Method of moments)



Note → When only three members come together at a joint and two of them are collinear then third member is a zero member. If no external force or support reaction is applied at that joint, it will be a zero member or compression.
 2) When two members (non-collinear) form a truss joint, both are zero force members if no external force or support reaction is applied at that joint.

Method of joints. In this method, each and every joint is treated as a free body diagram in equilibrium. The unknown forces are then determined by equilibrium equations $\sum F_x = 0$, $\sum F_y = 0$. While selecting a joint for calculation work, the joint should not contain more than two members in which the forces are unknown.

Example



Tension (+ve)
Compression (-ve)

In all cases, the joint analysis should start at a joint having at least one known force and at most two unknown forces.

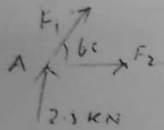
$$\sum F_x = 0, \sum F_y = 0$$

$\sum M_A =$

$$R_D \times 6 = 2 \times 1.5 + 4 \times 4.5 \Rightarrow R_D = 3.5 \text{ kN}$$

$$R_A = 2.5 \text{ kN}$$

For Point A



$$\sum V = 0$$

$$2.5 + F_1 \sin 60 = 0$$

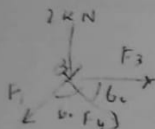
$$F_1 = -\frac{2.5}{\sin 60} = -2.887 \text{ N (C)}$$

$$\sum H = 0$$

$$F_2 + F_1 \cos 60 = 0$$

$$F_2 = -F_1 \cos 60 = -(-2.887) \cos 60 = 1.444 \text{ kN (Tension)}$$

For Point B



$$\sum V = 0$$

$$-F_1 \sin 60 + F_4 \sin 60 + 2 = 0$$

$$+F_4 \sin 60 = 2 - 2.887 \sin 60$$

$$\sin 60 \times [(-2.887) + F_4] = 2 - 2.887 \sin 60$$

$$F_4 = 0.577 \text{ kN (T)}$$

$$\sum H = 0$$

$$F_3 - F_1 \cos 60 + F_4 \cos 60 = 0$$

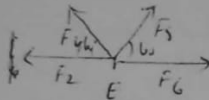
$$F_3 = F_1 \cos 60 - F_4 \cos 60$$

$$= (-2.887 \cos 60 - 0.577 \cos 60)$$

$$= -1.733 \text{ kN (C)}$$

$$F_4 = 0.577 \text{ kN (T)}$$

For Point E



$$\sum V = 0$$

$$F_5 \sin 60 + F_4 \sin 60 = 0$$

$$F_5 \sin 60 = -F_4 \sin 60 \Rightarrow F_5 = -F_4$$

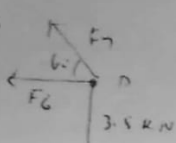
$$\sum H = 0$$

$$F_6 + F_5 \cos 60 = F_2 - F_4 \cos 60 = 0$$

$$F_6 + (-0.577) \cos 60 = 1.444 - 0.577 \cos 60 = 0$$

$$F_6 = 0.577 \cos 60 + 1.444 + 0.577 \cos 60 = 2.02 \text{ kN (T)}$$

For Point D



$$\sum V = 0$$

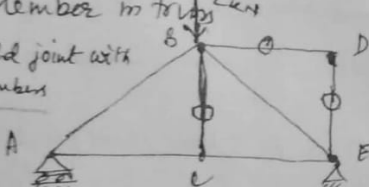
$$F_7 \sin 60 + 3.5 = 0$$

$$F_7 = -\frac{3.5 \sin 60}{\sin 60}$$

$$F_7 = -4.04 \text{ kN (C)}$$

Zero force Member in truss

regard loaded joint with two or more members



1) D joint has two member and no load or force, both members are zero force members (By inspection)

2) Joint C has three Members and no load or force, Since two member are collinear, other member (CB) is zero force member

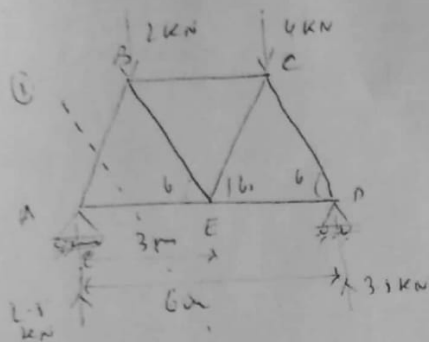
These members contribute to the stability of structure by providing buckling prevention for long slender members under compressive force

These Members can carry loads in the event that variations are introduced in the normal external loading configuration.

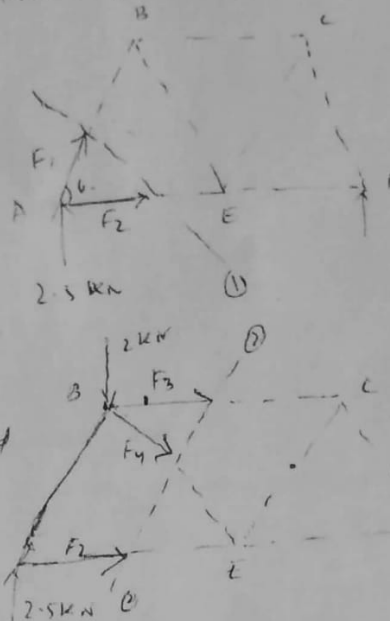
Method of sections: This method is particularly convenient when the forces in a few members of a truss are required to be found out. In this method, a section line is passed through the member or members in which forces are required to be found out. A part of structure on any side of section line, is then treated as a free body diagram in equilibrium with the action of external forces. The unknown forces are then found out by equilibrium.

$$\sum M = 0, \quad \sum F_x = 0, \quad \sum F_y = 0$$

Note: While drawing a section line, care should be taken not to cut more than 3 members in which forces are unknown.



In this method, the equilibrium of a portion of the truss is considered which is obtained by cutting the truss by an imaginary section.



Take left part of truss only
Taking Moment about point E
 $\sum M_E = 0$

$$2.5 \times 3 + F_1 \times 3 \sin 60^\circ = 0$$

$$F_1 = -2.887 \text{ (C)}$$

$$\sum F_x = 0$$

$$F_2 = -F_1 \cos 60^\circ$$

$$= -(-2.887) \times \frac{1}{2} = 1.44$$

$$\sum M_D = 0$$

$$F_3 \times 3 \sin 60^\circ + 2.5 \times 3 - 2 \times 1.5 = 0$$

$$3 F_3 \sin 60^\circ = -7.5 + 3$$

$$= -4.5$$

$$F_3 = \frac{-1.5 \times 2}{\sqrt{3}} = -1.73$$

For F_4

$$\sum M_A = 0$$

$$-2 \times 3 \cos 60^\circ - F_4 \times 3 \sin 60^\circ + F_3 \times 3 \sin 60^\circ = 0$$

$$3 F_4 \sin 60^\circ = -3 - (-1.732) \times 3 \times \frac{\sqrt{3}}{2}$$

$$F_4 = -2.45$$

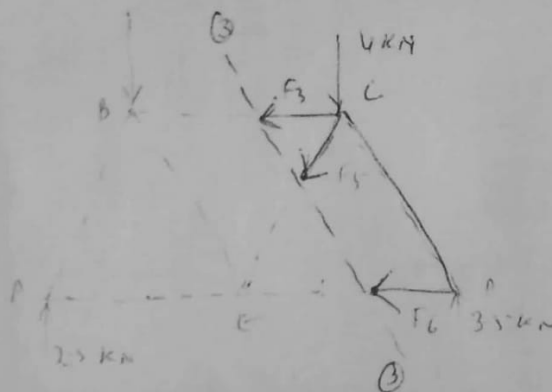
$$3 F_4 \sin 60^\circ = -3 + \frac{9}{2}$$

$$= -3 + 4.5$$

$$3 F_4 \sin 60^\circ = +1.5$$

$$F_4 \sin 60^\circ = +0.5$$

$$F_4 = \frac{0.5 \times 2}{\sqrt{3}} = 0.577 \text{ N}$$



$$\sum M_C = 0$$

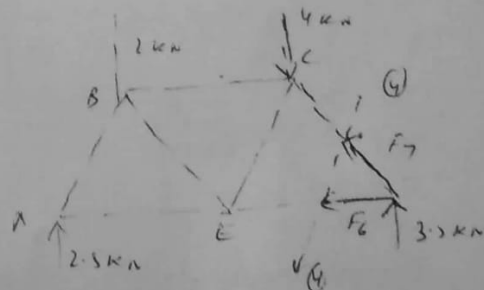
$$3.5 \times 3 \cos 60^\circ - F_6 \times 3 \sin 60^\circ = 0$$

$$F_6 = \frac{3.5 \times 3 \cos 60^\circ}{3 \sin 60^\circ} = 1.732 \text{ kN (T)}$$

$$\sum M_D = 0$$

$$F_5 \times 3 \sin 60^\circ + 4 \times 3 \cos 60^\circ + F_7 \times 3 \sin 60^\circ = 0$$

$$F_5 = -0.577 \text{ kN (C)}$$

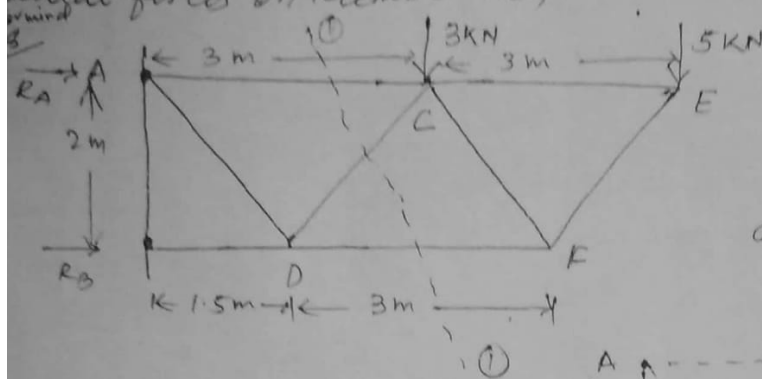


$$\sum M_E = 0$$

$$F_7 \times 3 \sin 60^\circ + 3.5 \times 3 - 2 \times 1.5 = 0$$

$$F_7 = -4.04 \text{ kN (C)}$$

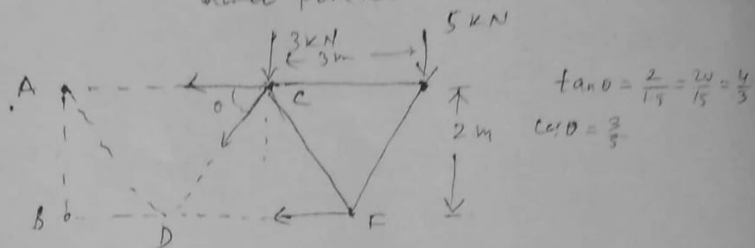
NO.1 A cantilever truss is loaded and supported as shown in fig. Find the axial forces on member AC, DC and DF.



In this case we need not determine the support reactions.

Pass section ①-① cutting members AC, DC and DF.

And consider the equilibrating right hand portion.



Taking Moment about D, $\sum M_D = 0$

$$F_{AC} \times 2 - 3 \times 1.5 - 5 \times 4.5 = 0$$

$$-2F_{AC} = 27 \Rightarrow F_{AC} = -13.5 \text{ kN (T)}$$

Take Moment about C, $\sum M_C = 0$

$$-5 \times 3 - F_{DF} \times 2 = 0 \Rightarrow F_{DF} = -7.5 \text{ kN (C)}$$

Take $\sum F_x = 0$ at joint C

$$-F_{AC} - F_{CD} \cos \theta - F_{DF} = 0$$

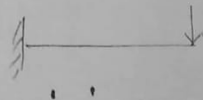
$$F_{CD} \cos \theta = -(F_{AC} + F_{DF})$$

$$= -(13.5 - 7.5)$$

$$F_{CD} \cos \theta = -6 \Rightarrow F_{CD} = \frac{-6}{\cos \theta} = \frac{-6}{3/5} = -10 \text{ kN (C)}$$

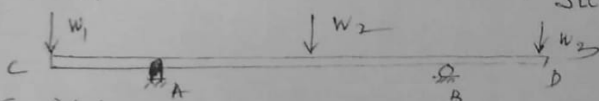
Types of beams:

- 1) Cantilever beam
One end fixed other end free
- 2) Simply Supported beam
Ends rest on walls, columns or knife edges



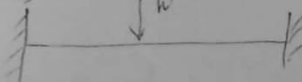
3) Overhanging beam

Supports are situated at the ends (one or both ends project beyond supports)

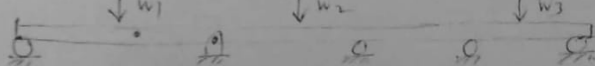


4) Fixed beam

(Both ends are fixed)

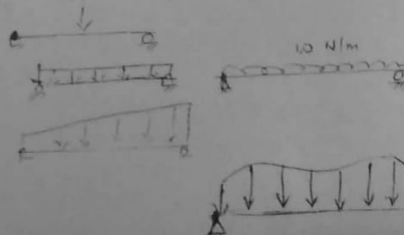


5) Continuous beam (A beam which has more than two supports)



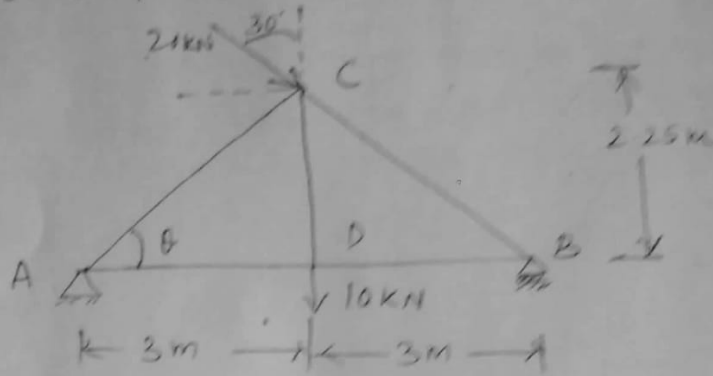
Types of loading

- 1) Concentrated or Point load
- 2) Uniformly distributed load
- 3) Uniformly Varying load
- 4) Varying load

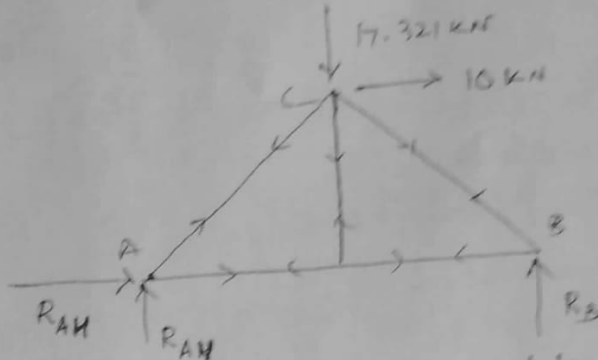


Ex. 6/1
S. Rajasekharan

Find the forces in all members of the truss as shown in fig.



Solution:



Reactions at A and B must be determined

$$\sum F_x = 0 \quad R_{AH} + 10 = 0 \Rightarrow R_{AH} = -10 \text{ kN}$$

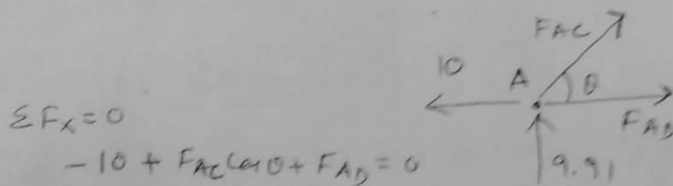
$$\sum F_y = 0 \quad R_{AV} + R_B - 10 - 17.321 = 0 \Rightarrow R_{AV} + R_B = 27.321$$

$$\odot \sum M_A = 0 \quad -17.321 \times (3) - 10(3) - 10(2.25) + R_B \times 6 = 0$$

$$R_{AV} = 9.910 \text{ kN}, \quad R_B = 17.411 \text{ kN}$$

$$\tan \theta = \frac{CD}{AD} = \frac{2.25}{3} \quad \theta = 36.87^\circ$$

FBD of joint A (Two members AD and AC are unknown)



$$\sum F_x = 0$$

$$-10 + F_{AC} \cos \theta + F_{AD} = 0$$

$$\sum F_y = 0$$

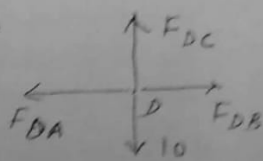
$$9.91 + F_{AC} \sin \theta = 0$$

$$\sin \theta = 0.6$$

$$\cos \theta = 0.8$$

$$F_{AC} = -16.517, \quad F_{AD} = 23.214 \checkmark$$

FBD of joint D



$$\sum F_x = 0$$

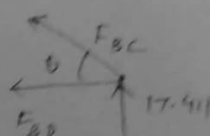
$$-F_{AD} + F_{DB} = 0$$

$$F_{DB} = F_{AD} = 23.214 \checkmark \text{ kN}$$

$$\sum F_y = 0$$

$$F_{DC} - 10 = 0 \Rightarrow F_{DC} = 10 \text{ kN} \checkmark$$

FBD of joint B



$$\sum F_x = 0$$

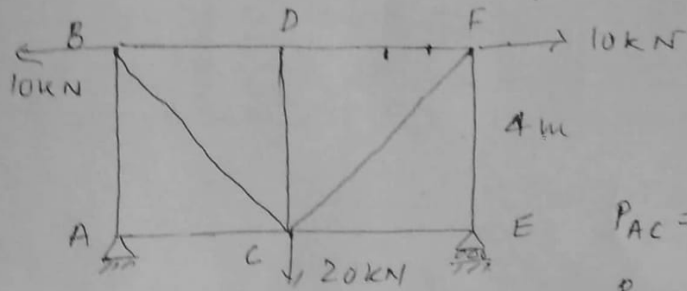
$$-F_{BD} - F_{BC} \cos \theta = 0$$

$$F_{BC} = -29.016 \checkmark$$

$$\sum F_y = 0$$

$$17.411 + F_{BC} \sin \theta = 0$$

Find the axial forces in all members of the truss as shown in fig.



$$P_{AC} = P_{CE} = 0$$

$$P_{AB} = -10$$

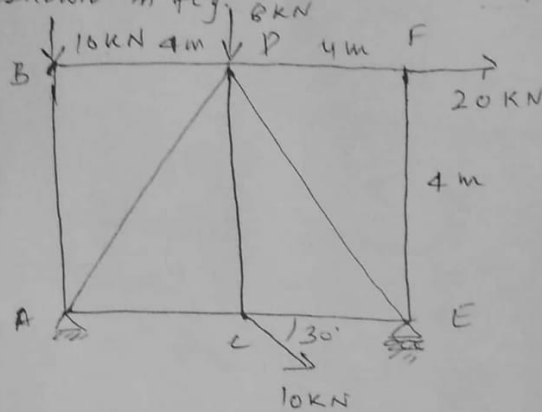
$$P_{BC} = 12.5 = P_{CF}$$

$$P_{EF} = -10$$

$$P_{BD} = P_{DF} = 2.5$$

$$P_{CD} = 0$$

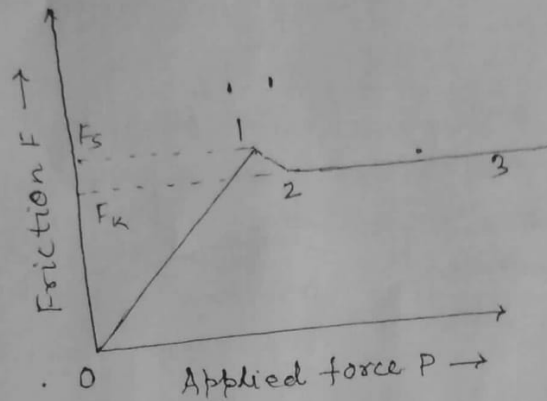
1) Find the forces in all the members of the truss as shown in fig.



FRICTION

Friction is a necessary evil, and needs to be mitigated.

Power screws, bearings and gears Flow of fluids in pipe, belt and rope, holding and fastening devices.



F_s (limiting friction)
static friction
 $F_k < F_s$
(dynamic) friction

Nature of friction:

- 1) Dry friction
 - Sliding friction
 - Rolling friction
- 2) Fluid friction

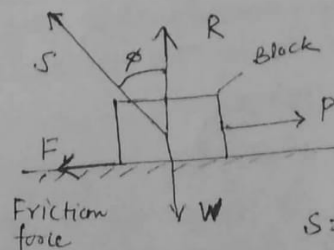
Laws of Dry friction :-

- 1) Friction acts tangential to the surface in contact and is in a direction opposite to that in which motion is to impend.
- 2) Friction force is maximum at the instant of impending motion
- 3) The Magnitude of limiting friction bears a constant ratio to the normal reaction between the mating surfaces.
- 4) Limiting friction is independent of the area and shape of the contact surface
- 5) Limiting friction depends upon the nature (roughness or smoothness) of the contact surfaces.

Angle of friction :-

$$\tan \phi = \frac{F}{R}$$

It is defined as the angle which the resultant of normal reaction and limiting friction makes with normal reaction

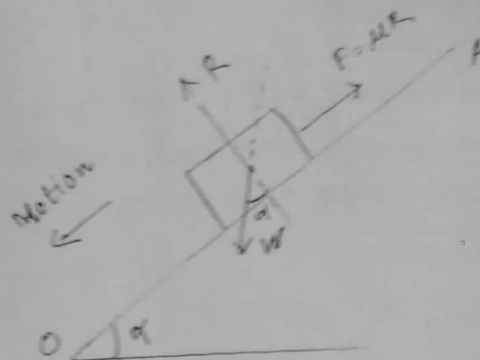


R = Normal reaction

S = Total resultant of F and R

The ratio of $\frac{F}{R}$ is also called Coefficient of friction (μ)

Angle of Repose:



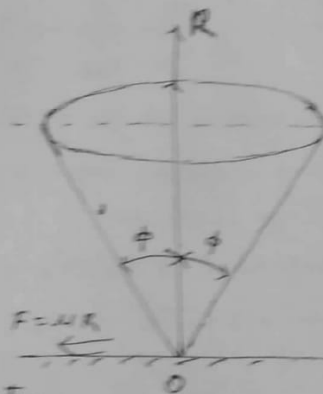
Consider a block of weight W resting on an inclined plane OA making an angle α with the horizontal. Let α be increased gradually till the block is just at the point of sliding

$$\left. \begin{aligned} \mu R &= W \sin \alpha \\ R &= W \cos \alpha \end{aligned} \right\} \begin{aligned} \mu &= \tan \alpha \\ \tan \phi &= \tan \alpha \\ \alpha &= \phi \end{aligned}$$

The angle α of the inclined plane at which the block resting on it is about to slide down the plane is called angle of repose and it is equal to angle of friction between the block and the inclined plane.

Cone of friction:

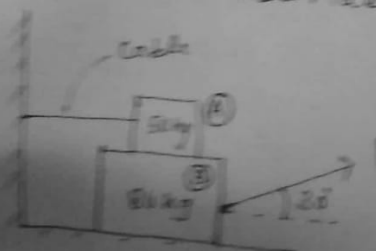
The lines of action of normal reaction R and friction force F meet at O . The magnitude of total reaction $S = \sqrt{F^2 + R^2}$ and it makes an angle ϕ with the normal reaction. O is the vertex of the cone whose axis is R and semivertex angle is ϕ . Such a right circular cone is called cone of friction.



When friction force is less than limiting friction the total reaction would lie within the cone. In that case the forces acting on the body are not sufficient to cause the motion. This aspect forms the working principle of self locking machines.

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No. 1. Two blocks A and B weighing 50 kg and 80 kg respectively are in equilibrium in the position as shown in fig. Calculate the force P required to move lower block B and tension in the cable. Take coeff. of friction at all contact surfaces to be 0.3



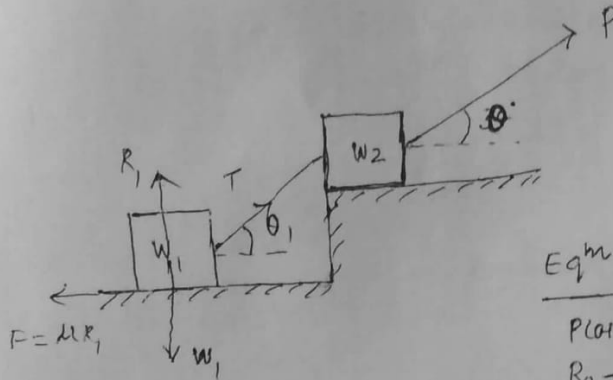
Solⁿ FBD of block A

$$\begin{aligned} \sum F_x = 0 &\Rightarrow -T + \mu R_1 = 0 \Rightarrow T = 0.3 R_1 \\ \sum F_y = 0 &\Rightarrow R_1 - W_1 = 0 \Rightarrow R_1 = W_1 = 490.5 \text{ N} \\ T &= 0.3 \times 490.5 = 147.15 \text{ N} \end{aligned}$$

FBD of block B, $\sum F_x = 0, \sum F_y = 0$

$$\boxed{P = 521.4 \text{ N}}$$

Q.NO.2: Two blocks of weight W_1 and W_2 are connected by a string and rest on a horizontal plane as shown in fig. Find the magnitude and the least force P that needs to be applied to the upper block to induce sliding. For each block and the block, the coefficient of friction may be taken as μ .



Eq^m of block W_1 ,

$$T \cos \theta_1 - \mu R_1 = 0 \quad \text{--- (1)}$$

$$T \sin \theta_1 + R_1 - W_1 = 0 \quad \text{--- (2)}$$

$$T = \frac{\mu W_1}{\cos \theta_1 + \mu \sin \theta_1} \quad \text{from (1) and (2)}$$

Eq^m of block W_2 :

$$P \cos \theta - T \cos \theta_1 - \mu R_2 = 0 \quad \text{--- (4)}$$

$$R_2 - W_2 + P \sin \theta - T \sin \theta_1 = 0 \quad \text{--- (5)}$$

$$T = \frac{P (\cos \theta + \mu \sin \theta) - \mu W_2}{\cos \theta_1 + \mu \sin \theta_1}$$

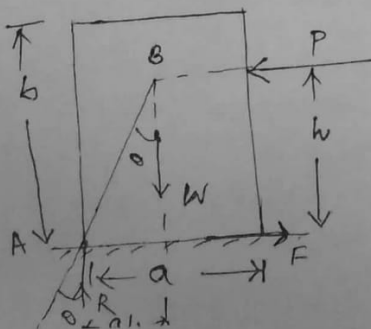
$$\text{So } \frac{\mu W_1}{\cos \theta_1 + \mu \sin \theta_1} = \frac{P (\cos \theta + \mu \sin \theta) - \mu W_2}{\cos \theta_1 + \mu \sin \theta_1}$$

$$\begin{aligned} P &= \frac{\mu (W_1 + W_2)}{\cos \theta + \mu \sin \theta} \\ &= \frac{\mu (W_1 + W_2)}{\cos \theta + \frac{\sin \phi}{\cos \phi} \sin \theta} \\ &= \frac{(W_1 + W_2) \sin \phi}{\cos (\theta - \phi)}, \end{aligned}$$

P will be least if the denominator is max^m.
 $\cos (\theta - \phi) = 1 \Rightarrow \theta = \phi$

$$P_{\min} = (W_1 + W_2) \sin \phi$$

Q.NO.3: A homogeneous block of weight W and height b rest on a horizontal plane on its square base of side a . A force P is applied horizontally to the block at height h . If the coeff. of friction is μ . Determine the maximum value of h so that block slides without toppling.

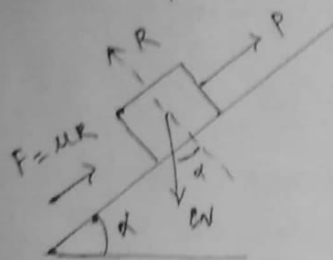


At the point of toppling entire reaction betⁿ the plane and the block will be at point A.

Resultant of F and R will pass through B.

$$\tan \theta = \frac{a/2}{h} \Rightarrow h = \frac{a}{2 \tan \theta} \Rightarrow h = \frac{a}{2 \mu}$$

2. No. 4: A body of weight w is placed on a rough inclined plane of inclination α . Find the force which should be applied parallel to the plane so that the body is on the point of motion i) down the plane 2) up the plane.



Resolving forces along and perpendicular to plane

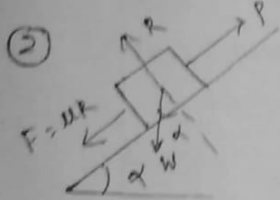
$$P + \mu R - W \sin \alpha = 0$$

$$R - W \cos \alpha = 0$$

$$P = W \sin \alpha - \mu W \cos \alpha$$

$$= W \sin \alpha - \frac{\sin \alpha + \mu \cos \alpha}{\cos \alpha}$$

$$P = \frac{W \sin(\alpha - \phi)}{\cos \phi}$$



$$\left. \begin{aligned} P - \mu R - W \sin \alpha &= 0 \\ R - W \cos \alpha &= 0 \end{aligned} \right\}$$

$$P = W \sin \alpha + \mu W \cos \alpha$$

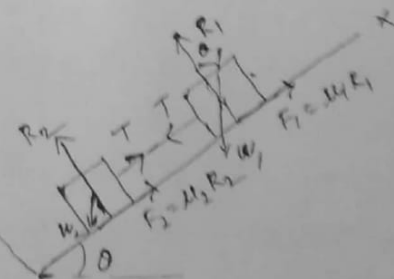
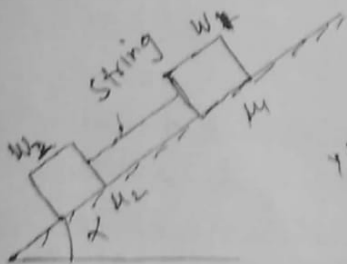
$$= W \sin \alpha + \frac{\sin \alpha + \mu \cos \alpha}{\cos \phi}$$

$$P = \frac{W \sin(\alpha + \phi)}{\cos \phi}$$

No. 5 Two blocks W_1 and W_2 rest on a rough inclined plane and are connected by a string. The coefficients of friction between these blocks and the plane are μ_1 and μ_2 respectively. Show that the blocks will be on the point of motion when the inclination of the plane with the horizontal is given by

$$\tan \theta = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

if $W_1 = W_2$ & $\mu_1 = \frac{1}{2}$, $\mu_2 = \frac{1}{3}$ what will be θ ?



Eqⁿ of W_1 , $\sum F_x = 0$ (along plane)

$$\mu_1 R_1 - T - W_1 \sin \theta = 0$$

$\sum F_y = 0$ ($\perp$ to plane)

$$R_1 - W_1 \cos \theta = 0$$

$$\mu_1 W_1 \cos \theta - W_1 \sin \theta - T = 0 \quad \text{--- (1)}$$

Eqⁿ of W_2

$$\mu_2 R_2 - W_2 \sin \theta + T = 0$$

$$R_2 - W_2 \cos \theta = 0$$

$$\mu_2 W_2 \cos \theta - W_2 \sin \theta + T = 0 \quad \text{--- (2)}$$

from (1) and (2)

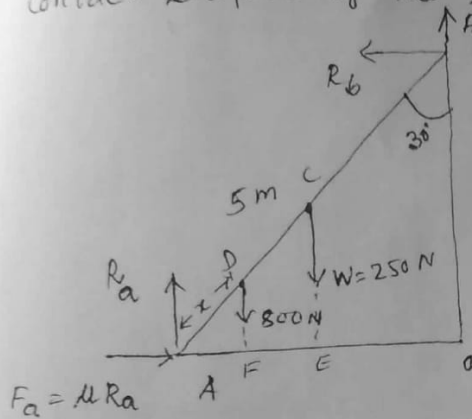
$$\mu_1 W_1 \cos \theta - W_1 \sin \theta = W_2 \sin \theta - \mu_2 W_2 \cos \theta$$

$$\tan \theta = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

$$\tan \theta = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

(Ladder friction)

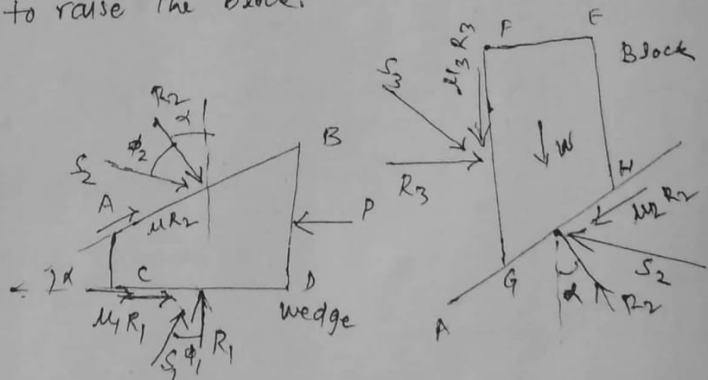
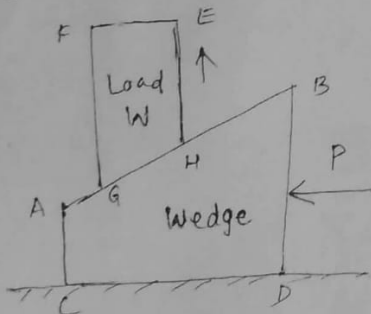
Q. NO. 1: A uniform ladder of weight 250 N and length 5 m is placed against a vertical wall in a position where its inclination to the vertical is 30° . A man weighing 800 N climbs the ladder. At what position will he induce slipping? Take coeff. of friction $\mu = 0.2$ at both the contact surfaces of the ladder.



$$\begin{aligned} \sum F_x &= 0, \\ R_b &= \mu R_a \Rightarrow R_a = \frac{R_b}{0.2} = 5R_b \\ R_a + \mu R_b &= 250 + 800 = 1050 \\ R_b &= 201.92 \text{ N} \\ \sum M_A &= 0 \\ R_b \times OB + \mu R_b \times OA &= 250 \times EA + 800 \times FA \quad \text{--- (1)} \\ OB &= 5 \cos 30, \quad OA = 5 \sin 30 \\ EA &= 2.5 \sin 30 \\ FA &= x \sin 30 \\ \text{from (1)} \quad x &= 1.66 \text{ m.} \end{aligned}$$

Wedge friction: $\rightarrow$

Q. NO. 2: A block weighing 2500 N, overlying a 10° degree wedge on a horizontal floor and leaning against a vertical wall, is to be raised by applying a horizontal force to the wedge. The coeff. of friction betⁿ all surfaces in contact is 0.3. Make calculations for the minimum horizontal force to be applied to raise the block.



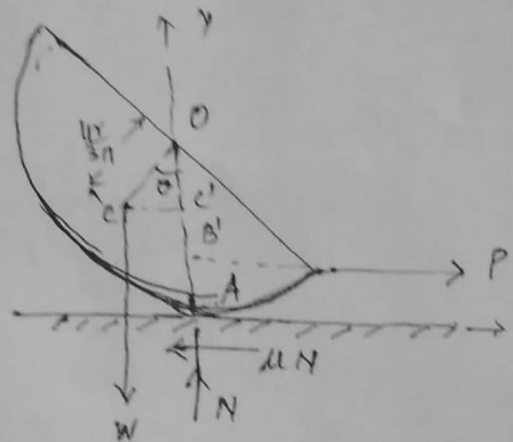
$$\mu = 0.3, \quad \phi = \tan^{-1}(0.3) = 16.7^\circ$$

E^m of block: $S_3 \cos \phi_3 - S_2 \sin(\alpha + \phi_2), \quad \phi_2 = \phi_3 = 16.7^\circ$
 $S_3 \cos 16.7 = S_2 \sin(10 + 16.7)$
 $S_2 = 2.132 S_3$

$$S_3 \sin \phi_3 + W - S_2 \cos(\alpha + \phi_2) = 0 \Rightarrow S_3 = 1545.8 \text{ N}, \quad S_2 = 3295.6 \text{ N}$$

E^m of wedge: $P - S_1 \sin \phi_1 - S_2 \sin(\alpha + \phi_2) = 0$
 $P - 0.2874 S_1 - 1480.8 = 0$
 $S_1 = 3074 \text{ N}$
 $P = 0.2874 S_1 + 1480.8$
 $= 2364.3 \text{ N.}$

Q.No: A short semicircular cylinder of radius r and weight W rests on a horizontal surface. It is pulled at right angles to its geometrical axis by a horizontal force P applied at the middle of the front edge. Find the angle θ that the flat surface will make with the horizontal plane just before sliding begins. Assume coeff. of friction at the point of contact as μ and all forces to be coplanar.



$$\sum F_y = 0$$

$$N - W = 0$$

$$N = W$$

$$\sum F_x = 0$$

$$P - \mu N = 0$$

$$P = \mu N$$

Taking Moment about A

$$W(CC') - P(AB') = 0$$

$$W(OC \sin \theta) - P(AO - B'O) = 0$$

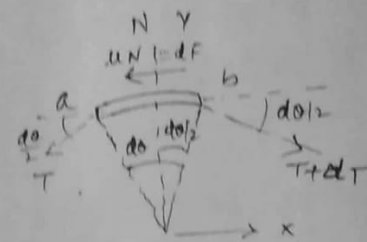
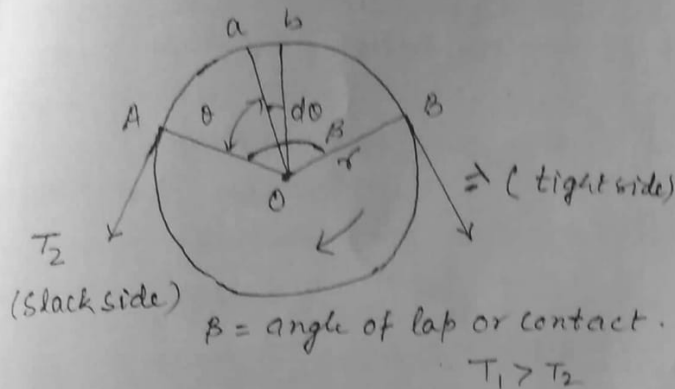
$$W\left(\frac{4r}{3\pi}\right) \sin \theta - P(r - r \sin \theta) = 0$$

$$W\left(\frac{4r}{3\pi}\right) \sin \theta - \mu W(r - r \sin \theta) = 0$$

$$\boxed{\sin \theta = \frac{3\mu\pi}{4 + 3\pi\mu}}$$

Belt Friction:

Ratio of Tensions (open belt)



Equation of equilibrium: $\sum F_x = 0$
 $T \cos \frac{d\theta}{2} + dF - (T + dT) \cos \frac{d\theta}{2} = 0 \quad \text{--- (1)}$

$\sum F_y = 0$
 $dN - T \sin \frac{d\theta}{2} - (T + dT) \sin \frac{d\theta}{2} = 0 \quad \text{--- (2)}$

As angle $d\theta$ is very small,
 $\cos \frac{d\theta}{2} = 1$, $\sin \frac{d\theta}{2} = \frac{d\theta}{2}$, $dF = \mu dN$

$T + \mu dN - (T + dT) = 0 \Rightarrow dT = \mu dN$

and $dN - T \frac{d\theta}{2} - (T + dT) \frac{d\theta}{2} = 0 \Rightarrow dN = T d\theta$ After neglecting dT
 $dT = \mu T d\theta$

$\frac{dT}{T} = \mu d\theta$

At A, $\theta = 0$, $T = T_2$
 at B, $\theta = \beta$, $T = T_1$

$\int_{T_2}^{T_1} \frac{dT}{T} = \mu \int_0^\beta d\theta \Rightarrow (\log_e T)_{T_2}^{T_1} = \mu \beta \Rightarrow \log_e \frac{T_1}{T_2} = \mu \beta$

$\boxed{\frac{T_1}{T_2} = e^{\mu \beta}}$

Ratio of tensions: (rope and V-belt drive)

$\frac{T_1}{T_2} = e^{(\mu \sec \alpha) \beta}$

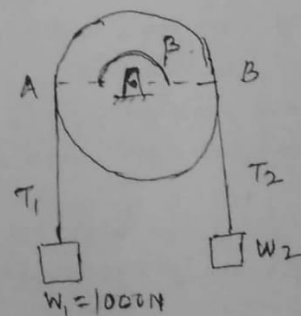
Q. NO. A belt supports two weight W_1 and W_2 over a pulley as shown in fig. If $W_1 = 1000 \text{ N}$, find the minimum weight W_2 to keep W_1 in equilibrium. Assume that pulley is locked and $\mu = 0.25$
 $T_1 > T_2$

Since W_2 just checks the tendency of W_1 to move down, Tension of side W_1 is larger

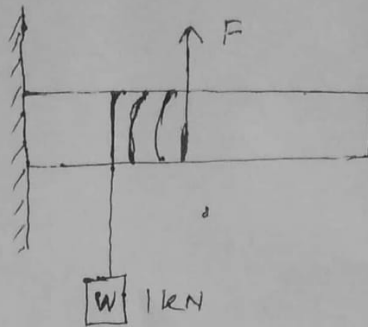
$\frac{T_1}{T_2} = e^{\mu \beta}$

$\mu = 0.25$
 $\beta = \pi$

$\frac{T_1}{T_2} = 2.718^{5(0.25)} \Rightarrow \frac{1000}{T_2} = 2.193$



Q.No. A rope is wrapped three turns around the cylinder as shown in fig. Determine the force required to just support a weight of 1 kN. Coeff. of friction between the rope and cylinder is 0.3.



$$\beta = 2\pi(3) = 6\pi$$

$$2\alpha = 180 \text{ (No groove)}$$

$$\alpha = 90$$

$$\frac{T_1}{T_2} = e^{\mu \cot \alpha \beta}$$

$$\frac{1000}{T_2} = e^{0.3 \times 1 \times 6\pi}$$

$$T_2 = \frac{1000}{285.5} = 3.5 \text{ N}$$

Transmission of Power through belt:

$$\text{Effective driving force} = (T_1 - T_2)$$

$$\text{Work done per second} = (T_1 - T_2) v$$

$$\text{Power transmitted} = \frac{(T_1 - T_2) v}{1000} \text{ kW}$$

Q.No: A belt is running over a pulley of diameter 120 cm at 200 r.p.m. The angle of contact is 165° and coefficient of friction between belt and pulley is 0.3. If the maximum tension in the belt is 3000 N. Find the power transmitted by the belt.

$$d = 1.2 \text{ m}, \quad N = 200 \text{ r.p.m.}$$

$$\theta = 165 \times \frac{\pi}{180}, \quad \mu = 0.3$$

$$T_1 = 3000 \text{ N}, \quad \frac{T_1}{T_2} = e^{0.3 \times 165\pi/180}$$

$$v = \frac{2\pi r N}{60} = \frac{\pi(1.2) \times 200}{60} = 12.56 \text{ m/s}$$

$$\frac{3000}{T_2} = e^{0.3 \times 165\pi/180}$$

$$T_2 = 1265 \text{ N}$$

$$\text{Power transmitted} = \frac{(T_1 - T_2) v}{1000} = \frac{(3000 - 1265) \times 12.56}{1000} = 21.79 \text{ kW}$$

Centre Points

Centroid, Centre of gravity, Centre of Mass.

Centre of gravity: It is the point through which resultant of the gravitational forces (weights) acts for any orientation of the body. Its position depends upon the shape of the body and may or may not be within the boundary of the body. It lies in a plane of symmetry, if any, of a body.

Centroid: The point where the entire length, area or volume is assumed to be concentrated is called the centroid. This point is called the centroid of line, centroid of area or centroid of volume as the case may be.

Centre of mass: The point where the entire mass of the body may be assumed to be concentrated.

The Centre of mass and centre of gravity and centroid are different only when gravitational field is not uniform and parallel and thickness, ρ 's and density is not uniform.

Centroid of one dimensional body.
$$\bar{x}_c = \frac{\int x dl}{\int dl}, \quad \bar{y}_c = \frac{\int y dl}{\int dl}$$

Centroid of two dimensional body
$$\bar{x}_c = \frac{\int x dA}{\int dA}, \quad \bar{y}_c = \frac{\int y dA}{\int dA}$$

To find the C.G. by Method of Moments.

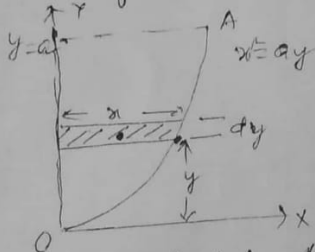
First Moment of area of a body about an axis through C.G. is zero. Why?

first moment of area w.r.t. axis

first moment of area w.r.t. x axis

Integration Method:

a) Choice of differential element



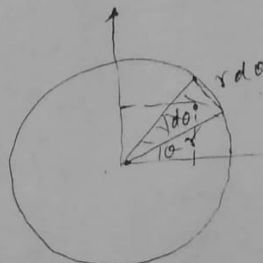
Area of differential element $x dy$
Position of its centroid $(\frac{x}{2}, y)$

Horizontal strip
Vertical strip
Triangular strip



Area of differential element $y dx$
Position of centroid $(x, \frac{y_a + y_b}{2})$

Triangular element



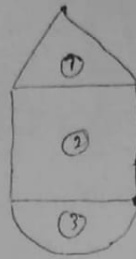
Area of element $dA = \frac{(r d\theta) r}{2}$
Position of centroid $(\frac{2}{3} r \cos \theta, \frac{2}{3} r \sin \theta)$

Choice of axis of reference: The determination of C.G. of a length, area or mass involve moments about some axis. So some frame of reference or coordinate axes are to be chosen. Though location of centroid does not depend upon reference axis chosen but the proper choice can considerably simplify the calculations. Axis of symmetry (if exists) if chosen as reference axis can simplify calculations.

If a figure or curve has some axis of symmetry, then centroid lies on the axis of symmetry. If a figure has more than one axis of symmetry, then intersection axis of symmetry is the centroid of given figure.

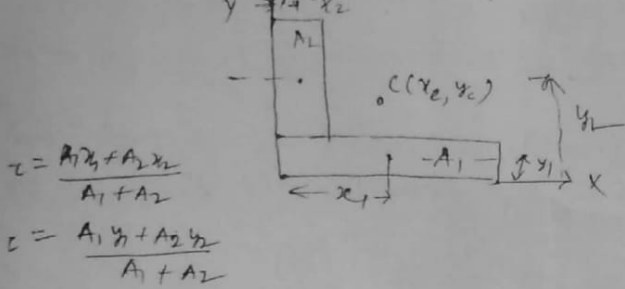
triod of composite plane figures.

Entire area can be considered to be made up of triangle, rectangle and a semicircle.



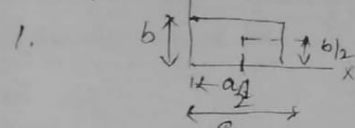
$$x_c = \frac{\sum \Delta A_i x_i}{\sum \Delta A_i} \quad y_c = \frac{\sum \Delta A_i y_i}{\sum \Delta A_i}$$

A Composite area is made of familiar figures whose position of centroid are known. If there is a hole or void (a non existing area), then it may be treated negative area.



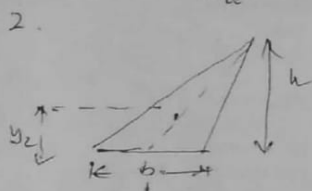
centroid of various shapes & areas

Rectangle



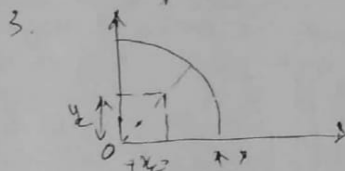
Area ab
 $x_c = \frac{a}{2}$
 $y_c = \frac{b}{2}$

Triangle



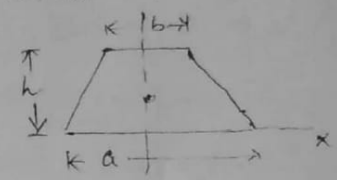
Area $\frac{bh}{2}$
 $x_c = \frac{b}{3}$
 $y_c = \frac{h}{3}$

Quarter circle



Area $\frac{\pi r^2}{4}$
 $x_c = \frac{4r}{3\pi}$
 $y_c = \frac{4r}{3\pi}$

Trapezium

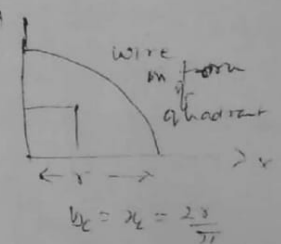


$y_c = \frac{h}{3} \left[\frac{a+2b}{a+b} \right], x_c = 0$

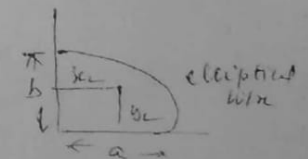
Semi circle



Area $\frac{\pi r^2}{2}$
 $x_c = 0$
 $y_c = \frac{4r}{3\pi}$

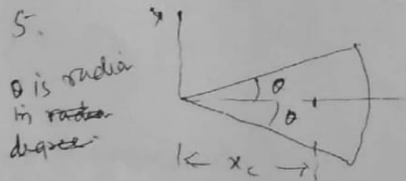


$y_c = x_c = \frac{2r}{3}$



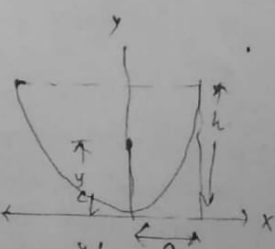
$x_c = \frac{4r}{3\pi}$
 $y_c = \frac{4r}{3\pi}$

Circular sector

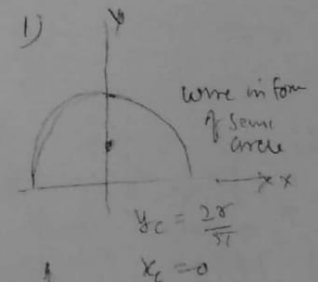


Area $r^2 \theta$
 $x_c = \frac{4r \sin \theta}{3\theta}$
 $y_c = 0$

Parabola

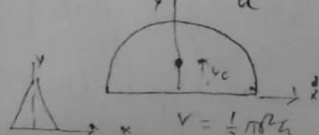


Area $\frac{4ah}{3}$
 $x_c = \frac{a}{2}$
 $y_c = \frac{3h}{5}$

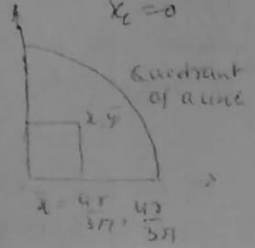


$y_c = \frac{2r}{3}$
 $x_c = 0$

Semicircle

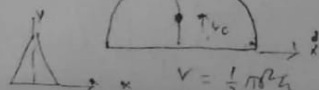


Volume $\frac{2}{3} \pi r^3$
 $x_c = 0$
 $y_c = \frac{4r}{3\pi}$



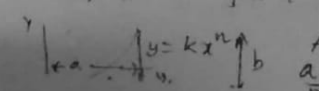
$x_c = \frac{4r}{3\pi}, y_c = \frac{4r}{3\pi}$

Right circular cone



Area $\frac{ab}{n+1}$
 $x_c = \frac{a}{n+1}$
 $y_c = \frac{b}{n+1}$

Conical spool



centroid of a wire bent in form of a quadrant of the circle of radius r

$$y_c = \frac{\int y dL}{\int dL} \quad \text{--- (1)}$$

eqⁿ of circle $x^2 + y^2 = r^2$

Diff. $2x dx + 2y dy = 0 \Rightarrow dy = -\frac{x}{y} dx$

$$dL = \sqrt{dx^2 + dy^2}$$

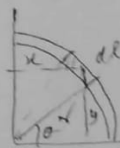
$$dL = \frac{r}{y} dx$$

Substituting in (1)

$$y_c = \frac{\int_0^r y \frac{r}{y} dx}{\int_0^r \frac{r}{y} dx} = \frac{\int_0^r r dx}{\frac{2\pi r}{4}} = \frac{r^2}{\frac{2\pi r}{4}} \Rightarrow y_c = \frac{2r}{\pi}$$

Because of symmetry $x_c = y_c = \frac{2r}{\pi}$

Let us solve this by using polar coordinates



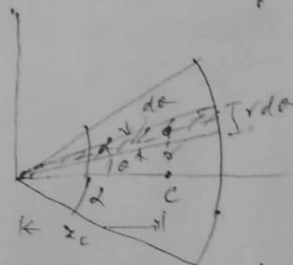
$$y_c = \frac{\int y dL}{\int dL}$$

$$y = r \sin \theta$$

$$dL = r d\theta$$

$$\frac{\int_0^{\pi/2} (r \sin \theta) r d\theta}{\int_0^{\pi/2} r d\theta} = \frac{2r}{\pi}$$

Determine coordinate of C.G. of Lamina in shape of circular sector of radius r and central angle α



$$x_c = \frac{\int x dA}{\int dA}$$

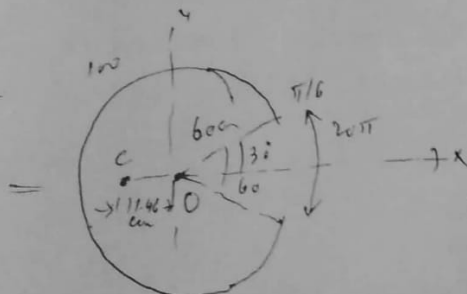
$$dA = \frac{r^2 d\theta}{2}$$

$$= \frac{\int_0^\alpha \left(\frac{2}{3} r \cos \theta\right) \frac{r^2 d\theta}{2}}{\int_0^\alpha \frac{r^2 d\theta}{2}} = \frac{2r}{3} \left(\frac{\sin \alpha}{\alpha}\right)$$

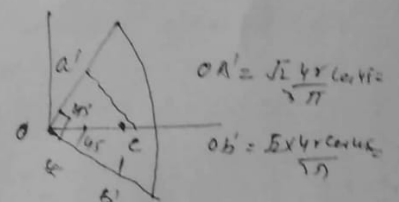
Centroid of a quadrant of a circle can be determined $2\alpha = \pi/2, \alpha = \pi/4$

$$x_c = \frac{2r \sin \pi/4}{3 \pi/4} = \frac{\sqrt{2}}{3} \frac{4r}{\pi}$$

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1st term
2008



$$x_c = \frac{\int_0^{11\pi/6} (r \cos \theta) r d\theta}{\int_0^{11\pi/6} r d\theta}$$



$$OA' = \frac{\sqrt{2}}{3} \frac{4r}{\pi}$$

$$OB' = \frac{2r \sin \alpha}{3\alpha}$$

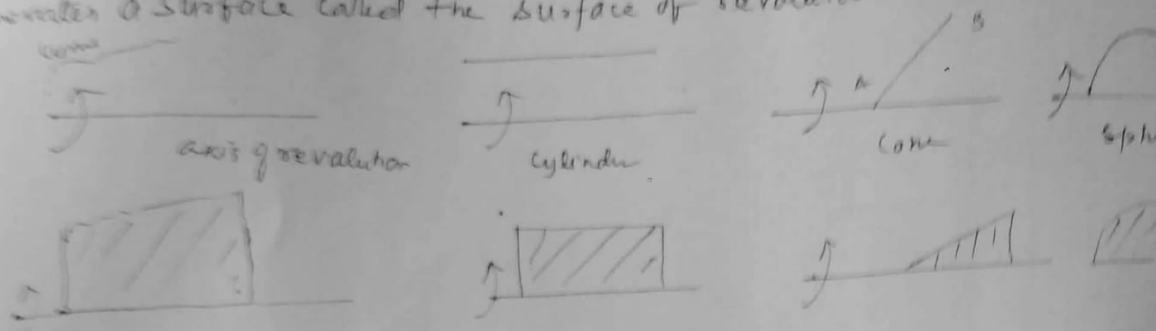
$$= \frac{60^2 (\sin \theta)_{11\pi/6}}{60 \left(\frac{11\pi}{6}\right)} = \frac{36}{\pi} = 11.4$$

Alternative Method

$$x_c = \frac{12 \pi (1) - 20 \pi \left(\frac{2r \sin \alpha}{\alpha}\right)}{100 \pi} = 11.46 \text{ cm}$$

where $\alpha = 30^\circ$

Pappus-Guldinus theorems: When a plane curve rotates about a fixed axis, it generates a surface called the surface of revolution.



A solid cylinder may be generated by rotating a rectangular area, a cone by rotating a triangular area, a solid sphere by rotating a semi-circular area.

Theorem 1: The area of surface generated by rotating a plane curve about a non-intersecting axis in its plane is equal to the product of the length of the curve and distance travelled by its centroid.



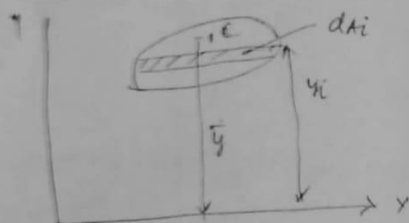
$$\bar{y} = \frac{\int y_i di}{\int di}$$

$$\int 2\pi y_i di$$

$$= l \times 2\pi \bar{y}$$

$$= \text{length of generating curve} \times \text{distance travelled by its centroid}$$

Theorem 2: The volume of the solid generated by rotating any plane figure about a non-intersecting axis in its plane is equal to the product of the area of figure and the distance travelled by its centroid.



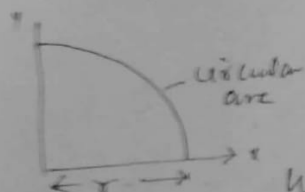
Volume generated by elementary area = $2\pi y_i di$

Volume generated by entire area = $\int 2\pi y_i di$

$$V = A \times 2\pi \bar{y}$$

Utility To compute areas of surfaces of revolution and volumes of bodies of revolution.

Ex. Using Pappus-Guldinus theorem, 1) Calculate the area of surface of revolution by revolving a quarter circular arc of rad r about $O-y$ axis. 2) Volume generated revolving a quarter ellipse about $O-x$.



Length of revolving arc = $\pi r/2$

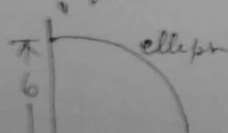
Distance of the centroid from y axis = $\frac{2r}{3}$

Distance travelled by centroid = $2\pi(\frac{2r}{3}) = 4\pi r$

Using Pappus-Guldinus theorem,

$$\text{Area generated} = \frac{\pi r}{2} \times 4\pi r = 2\pi^2 r^2$$

(ii) For a quarter ellipse

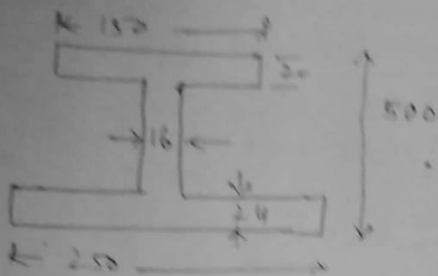


Distance of centroid from x axis = $\frac{4b}{3\pi}$

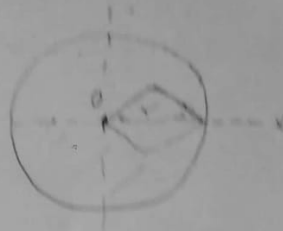
Distance travelled by centroid = $2\pi(\frac{4b}{3\pi}) = \frac{8b}{3}$

$$\text{Area} = \frac{\pi ab}{4}$$

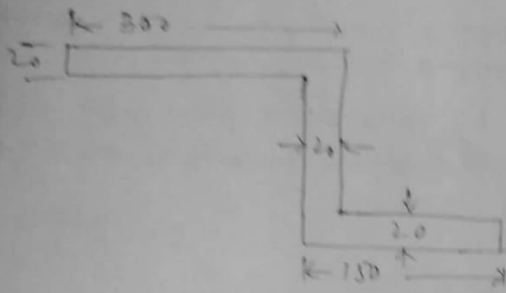
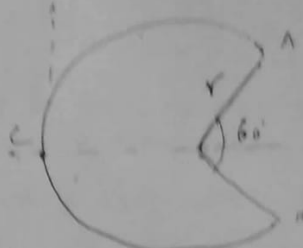
Area of ellipse = πab



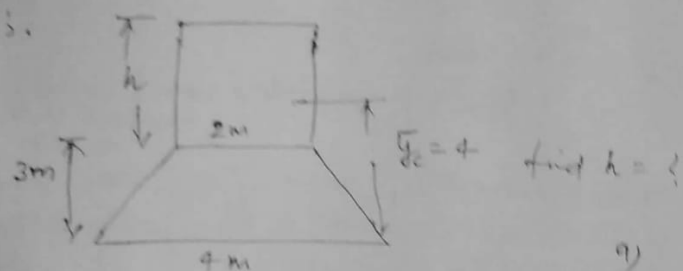
6)



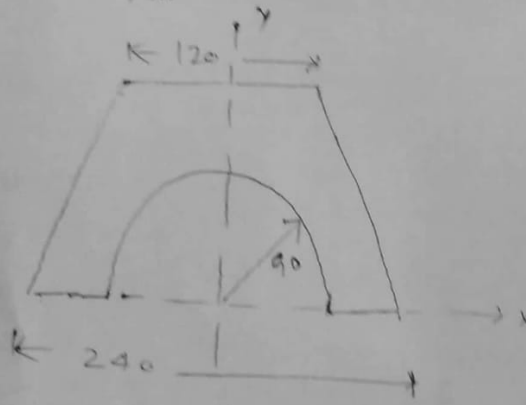
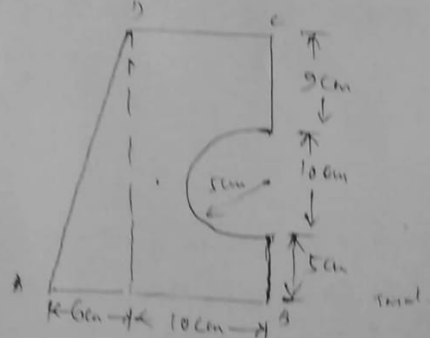
7)



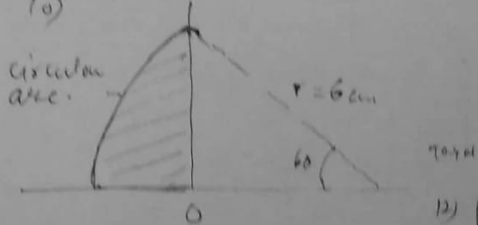
8)



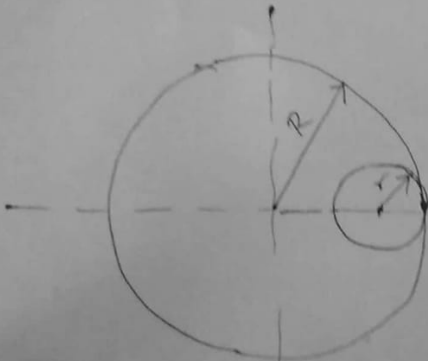
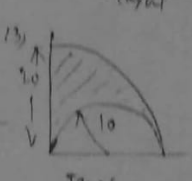
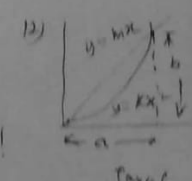
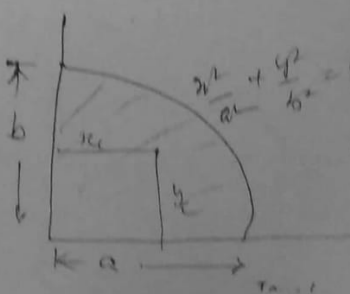
9)



10)



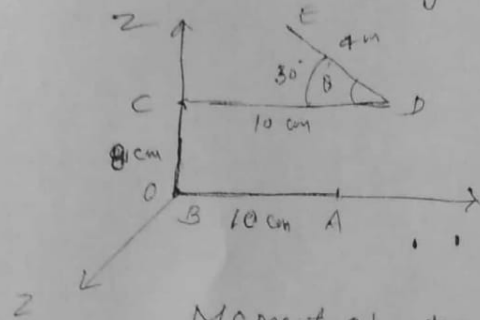
11)



$$\bar{A} \bar{x} = A_1 \bar{x}_1 - A_2 \bar{x}_2$$

$$\bar{x} = \frac{R^3 - 0^3}{3}$$

Centroid of a wire bent in a given shape.



$$\bar{x} = 4.15 \text{ cm}$$

$$\bar{y} = 4.62 \text{ cm}$$

$$L = \int dL = \sum L_i$$

Moment about x axis is given by

$$L\bar{y} = \int dL \cdot y = \sum L_i y_i$$

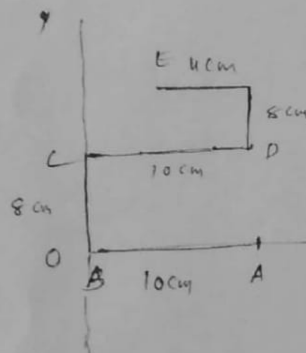
$$\bar{y} = \frac{\sum L_i y_i}{L}$$

Moment about y axis

$$L\bar{x} = \int dL \cdot x = \sum L_i x_i$$

$$\bar{x} = \frac{\sum L_i x_i}{L}$$

Find the C.G of the bent up wire as shown in fig.



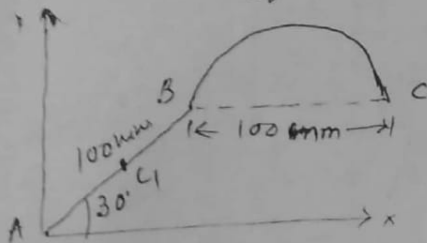
$$\begin{aligned} L_1 &= 10, \bar{x}_1 = 5, \bar{y}_1 = 0 \\ L_2 &= 8, \bar{x}_2 = 0, \bar{y}_2 = 4 \\ L_3 &= 10, \bar{x}_3 = 5, \bar{y}_3 = 8 \\ L_4 &= 4, \bar{x}_4 = (10+2) = 12, \bar{y}_4 = 16 \end{aligned}$$

$$\bar{x} = \frac{L_1 \bar{x}_1 + L_2 \bar{x}_2 + L_3 \bar{x}_3 + L_4 \bar{x}_4}{L_1 + L_2 + L_3 + L_4} = 5.3$$

$$\bar{y} = \frac{L_1 \bar{y}_1 + L_2 \bar{y}_2 + L_3 \bar{y}_3 + L_4 \bar{y}_4}{L_1 + L_2 + L_3 + L_4} = 6.8$$

Q.No.

Locate the centroid of the bent wire shown in fig-



Length Part 100 AB (L_1)
" " BC (L_2)

C_1 is at the centre of line AB

$$\begin{aligned} L_1 &= 100, \bar{x}_1 = 50 \cos 30 = 43.3 \\ \bar{y}_1 &= 50 \sin 30 = 25 \end{aligned}$$

$$L_2 = \pi r = \pi(50) = 157$$

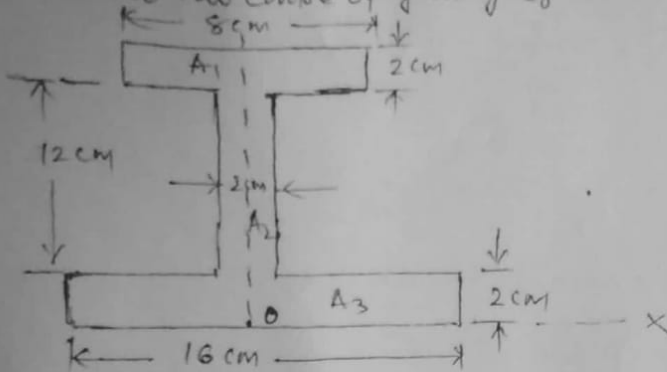
$$\bar{x}_2 = 100 \cos 30 + 50 = 136.6$$

$$\begin{aligned} \bar{y}_2 &= 100 \sin 30 + \frac{2r}{\pi} \\ &= 100 \sin 30 + \frac{2(50)}{\pi} = 81.83 \end{aligned}$$

$$\bar{x} = \frac{L_1 \bar{x}_1 + L_2 \bar{x}_2}{L_1 + L_2} = 100.3$$

$$\bar{y} = \frac{L_1 \bar{y}_1 + L_2 \bar{y}_2}{L_1 + L_2} = 59.72$$

Not. Find the centre of gravity of the I-section as shown in fig.

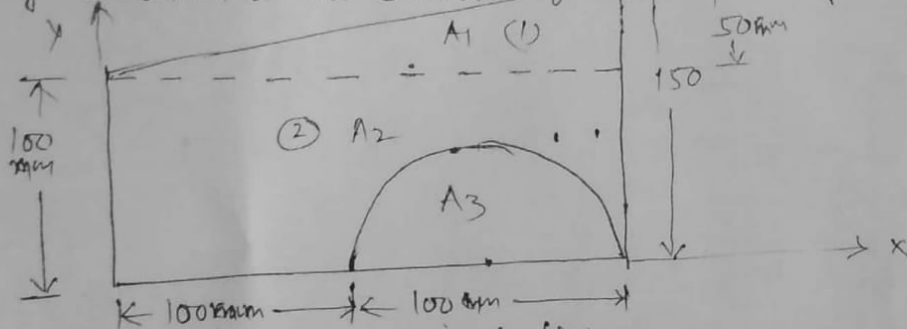


Let O be the origin $\bar{x} = 0$, (Symmetric about y-axis)

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + A_3 \bar{y}_3}{A_1 + A_2 + A_3} = \frac{16 \times 15 + 12 \times 2(8) + (16 \times 2)(1)}{(8 \times 2) + (12 \times 2) + (16 \times 2)} = 6.44 \text{ cm}$$

$$(\bar{x}, \bar{y}) = (0, 6.44)$$

Q.2: A Semicircular area is removed from the trapezoid as shown in fig. Determine the centroid of the remaining area



Divide the fig. into three simple figures

$$A_1 = \frac{1}{2} \times 50 \times 200, \quad \bar{x}_1 = \frac{2}{3} \times 200 = 133.34, \quad \bar{y}_1 = 150 + \frac{50}{3} = 166.67$$

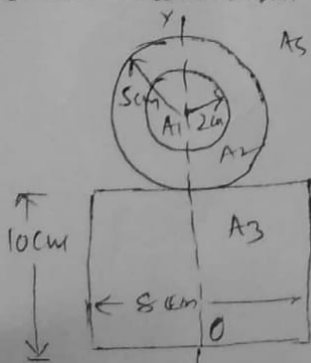
$$A_2 = 200 \times 100, \quad \bar{x}_2 = 100, \quad \bar{y}_2 = 50$$

$$A_3 = \frac{1}{2} \pi (50)^2, \quad \bar{x}_3 = 100 + 50 = 150, \quad \bar{y}_3 = \frac{48}{3\pi} = \frac{4 \times 50}{3\pi} = 21.22$$

$$\bar{x} = \frac{A_1 \bar{x}_1 + A_2 \bar{x}_2 + A_3 \bar{x}_3}{A_1 + A_2 + A_3} = 98.59 \text{ mm}$$

$$\bar{y} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + A_3 \bar{y}_3}{A_1 + A_2 + A_3} = 71.18 \text{ mm}$$

Ex.) Locate the centroid of the plane area as shown in fig.



As Symmetric about y axis $\bar{x} = 0$

$$A_1 = \pi (2)^2 = 12.57, \quad \bar{y}_1 = 15$$

$$A_2 = \pi (5)^2 = 78.54, \quad \bar{y}_2 = 15$$

$$A_3 = (10 \times 8) = 80, \quad \bar{y}_3 = 5$$

$$\bar{y} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + A_3 \bar{y}_3}{A_1 + A_2 + A_3} = 9.52$$

$$\text{Hence } (\bar{x}, \bar{y}) = (0, 9.52)$$